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Offshore Refining Crude Oil into Petroleum Products by Using Electromagnetic Radiation Waves at the Resonance Frequencies of Petroleum Products

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Research Abstract:

This research presents an innovative technique for crude oil refining using electromagnetic waves at resonance frequencies of petroleum products, offering a cleaner and more efficient alternative to conventional thermal distillation.

The method significantly reduces energy consumption and carbon emissions, with a compact modular design suitable for offshore refineries, and can be seamlessly integrated into existing infrastructure.

This approach represents a step toward sustainable and efficient refining, aligned with global decarbonization efforts.

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Significant advantages due to applying this method in offshore refinery crude oil:

1. Energy Efficiency: Reduction of Approximately 45% in Distillation Heat Energy Reduction in Carbon Emissions During the Refining Process: From 18% Up to 40%.
2. Offshore Refining Stations alongside Extraction Platforms Compact Refining Facilities for Smaller Geographic Areas
3. Reduction of High Construction Costs Associated with Vertical Distillation Towers
4. Easy Capacity Expansion with Horizontal Refining Units
5. Improved Workplace Safety with Reduced Refining Unit Heights
6. Quick Production and Assembly of Refining Units Using Modular Systems

For detail of advantage clarification refer to [Appendix A](#) Page No. (39).

Achievements of that research:

In my research and based on laboratory experiments:

- I. Conducted demonstrated the resonance frequencies for Naphtha, kerosene, and Diesel fuel.

- II. Establishes a scientific method for identifying any resonance frequency of any petroleum product.
- III. Based on my proposed method that selective heating to extract the petroleum products the comparison study with distillation tower refine ring method shows the reduction in heating energy by 45% and corresponding reduction in carbon emission by 18% up to 40%.

The next phase to be carried out to complete the research includes the following:

1. **Developing a prototype** for refining unit using magnetrons at the identified resonance frequencies to complete the study of production rates.
2. **Cost Analysis**
 - o Comprehensive comparison between the proposed refining method and conventional distillation.
 - o Evaluation includes **construction, operation, and maintenance** costs.
 - o Scalability assessed for increased production capacity.
3. **Future Recommendations:**
 - o Validate large-scale feasibility through pilot projects.
 - o Further improve **energy efficiency**.
 - o Optimize **resonance frequency targeting** for various petroleum products and various crude oil (Light -medium - heavy).
4. **Integration with existing refinery operations to accelerate industrial adoption.**

Innovation

The current process of refining crude oil in existing refineries:

Involves separating petroleum products by heated the Crude oil in a furnace and then sent into a distillation tower. As the heated crude oil rises up the column, various components, like gases, gasoline, kerosene, diesel, and heavy oils, condense at different heights of the column depending on their boiling points.

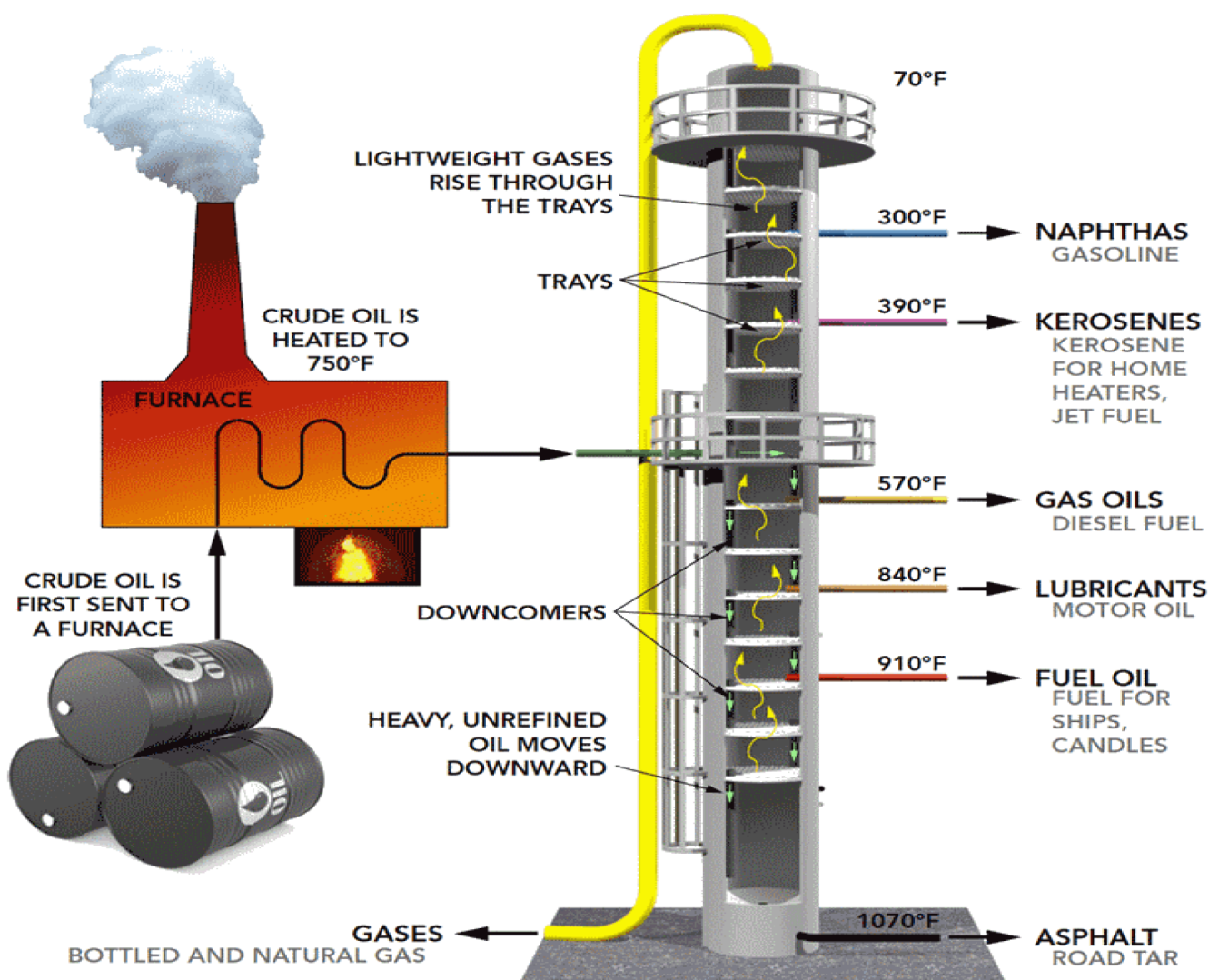


Figure 1.1—Illustration of a Distillation Tower

Supporting Literature Review

The proposed methodology of using electromagnetic radiation at the resonance frequencies of petroleum products is supported by previous studies in applied physics, industrial chemistry, and electromagnetic heating technologies.

Dunia and Edgar (2012) demonstrated the feasibility of using electromagnetic heaters to reduce the viscosity of heavy crude oil in pipelines, thereby improving flow efficiency and providing evidence that selective energy transfer through resonance can be effectively applied in petroleum processing. Similarly, studies on *microwave treatment in oil refining* (Applied Petrochemical Research, 2012) have shown the potential of resonant microwave cavities to efficiently convert electromagnetic energy into heat, which can be tailored for crude oil components.

Recent research by Jamaloei (2022) presented experimental and pilot-scale investigations on electromagnetic heating for heavy oil and bitumen recovery, highlighting both the efficiency and scalability of the technology for industrial use. Furthermore, Chhetri and Islam (2008) provided a critical review of electromagnetic heating for enhanced oil recovery, emphasizing its advantages over conventional thermal methods and its relevance for sustainable refining technologies.

These studies collectively provide strong evidence that electromagnetic resonance heating is a viable and innovative alternative to conventional distillation. They also reinforce the novelty of the present research in applying resonance-based separation specifically for crude oil refining into petroleum products.

Key Highlights:. Resonance Frequencies:

Laboratory experiments were conducted to identify the resonance frequencies of key petroleum products, including naphtha, kerosene, and diesel fuel. This step was carried out by me as part of this research.

Sequential Separation Process:

- **Crude oil is initially heated in furnaces to a temperature below its evaporation point.**
 - The preheated crude oil is transferred to the first horizontal unit in a series of refining units. These units are arranged in sequence based on the evaporation temperatures of the petroleum products (e.g., Naphtha → Kerosene → Diesel fuel, etc.).
 - In each unit, electromagnetic waves at the specific resonance frequency of the target product are applied.
 - The target product selectively absorbs these waves, converting the energy into heat and evaporating exclusively.

Condensation and Collection:

- The evaporated product is condensed at the top of its respective unit, allowing direct collection (Naphtha).
- The residual crude oil is then transferred to the next unit for the extraction of the subsequent product (kerosene in the second unit).
- This process is repeated for each unit until all desired petroleum products are extracted.

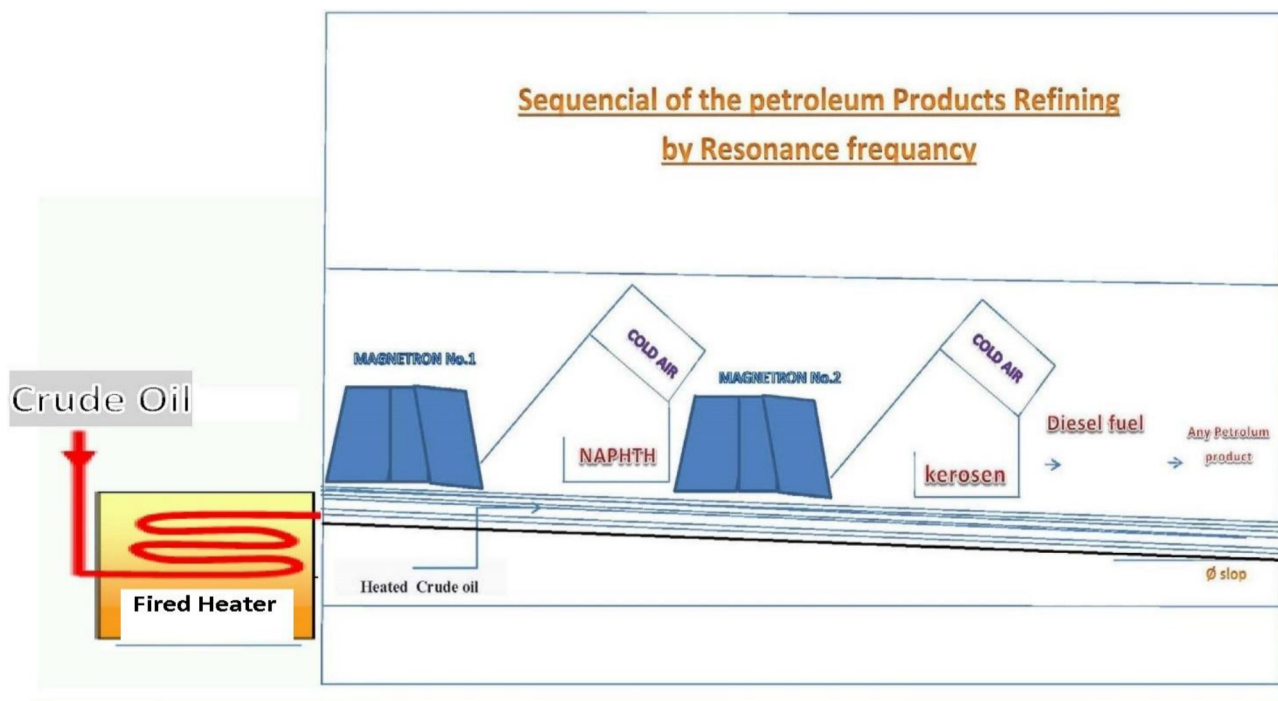


Figure 1.2—Illustration of Electromagnetic Wave Refinery Unit

Source of Electromagnetic Radiation: Magnetron

Magnetron units, commonly used in household microwave ovens and military applications, Magnetrons are manufactured globally by numerous factories, with capacities from 1 kW to 5500 kW.

These units are specifically designed to generate specific frequency needed, in our research Magnetron will be designed to generate electromagnetic waves with resonant frequencies tailored to each petroleum product. In production units,

The refinery's capacity is determined by the total number of Magnetrons installed in each petroleum product refining unit as well as their individual power outputs.

For detail of construction and efficiency of types of magnetron refer to [Appendix A](#) page No. (39).

Experimental Test to Identify the resonance frequencies of Naphtha, Kerosene and Diesel fuel.

This study involves identifying the resonance frequency of various petroleum products in the laboratory **Devices and Instrument Used in Laboratory Test:**

- A variable frequency pulse generator with a frequency range of 35 to 4400 MHz.
- A microcontroller board equipped with a digital thermal sensor.
- Excel software with the "PLX-DAQ-v2" add-in for data logging and analysis.
- A laptop with software for recording and plotting graphs of frequency versus sample temperature and sample temperature versus time.

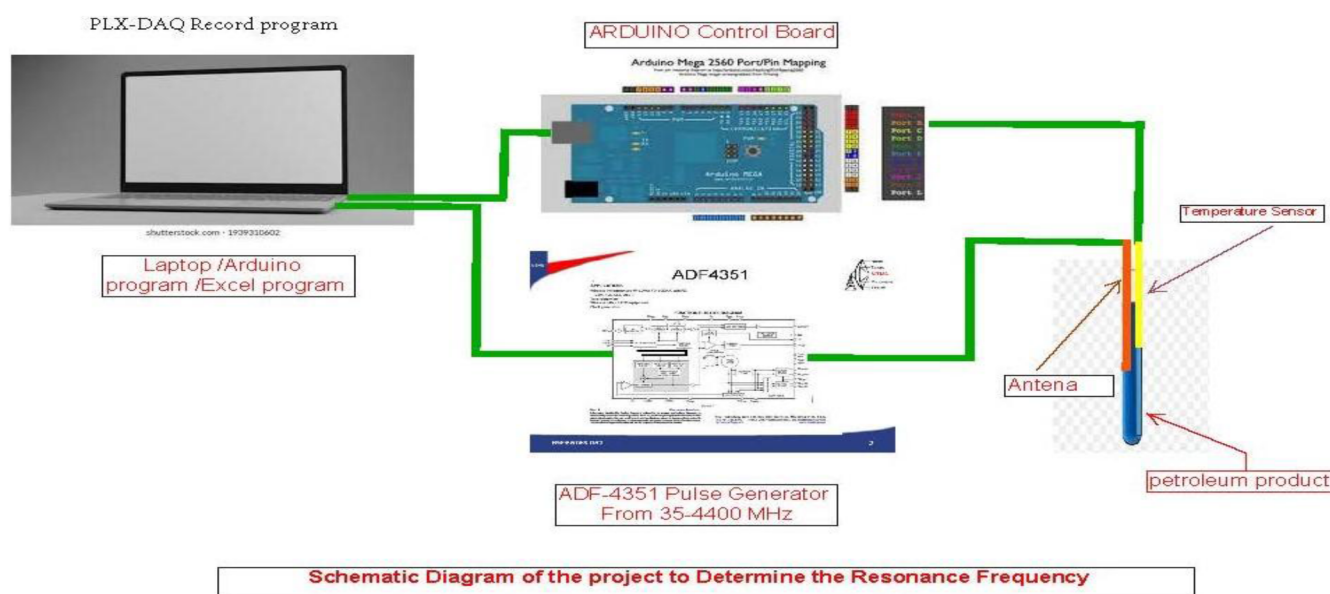


Fig. 1.6—The Research system for generate pulse, record and Analysis.

Steps for identifying any petroleum product resonance frequency for Example "Naphtha: ".

- **Experimental Setup:**

A pulse generator was used to produce electromagnetic radiation ranging from 35 MHz to 4400 MHz.

- A 14 cm³ sample of Naphtha was exposed to this radiation, starting at a frequency of 350 MHz and incrementing by 1 MHz every 10 seconds.
- **Temperature Measurement:**
- The sample's temperature was recorded every 10 seconds during the exposure.

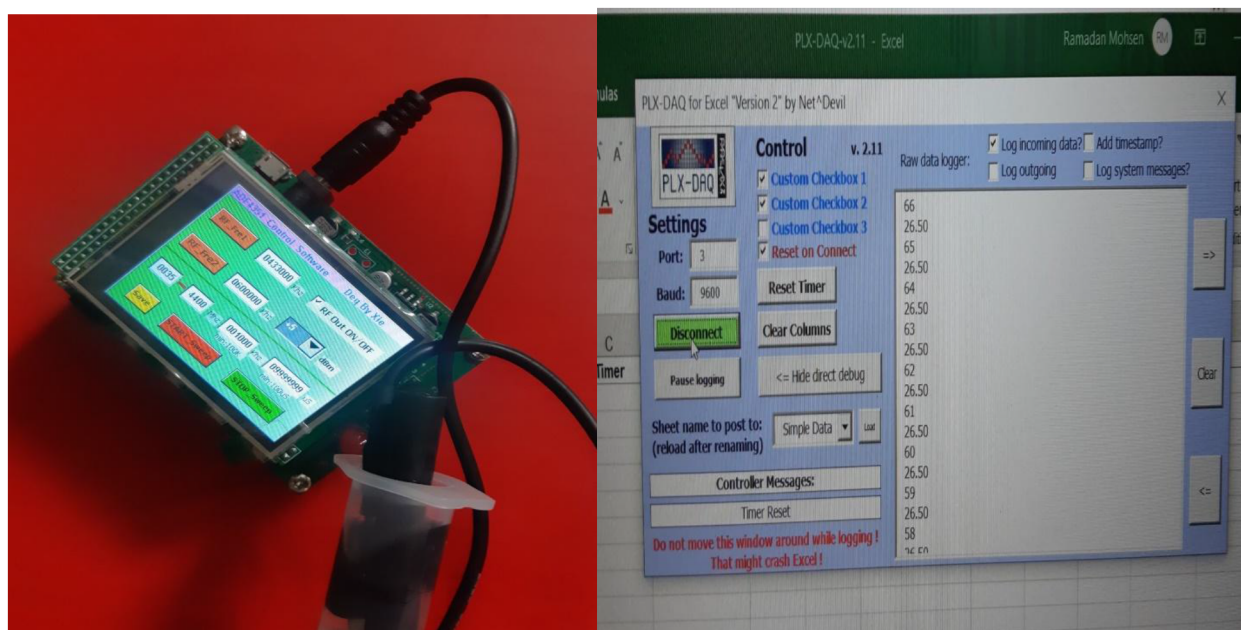
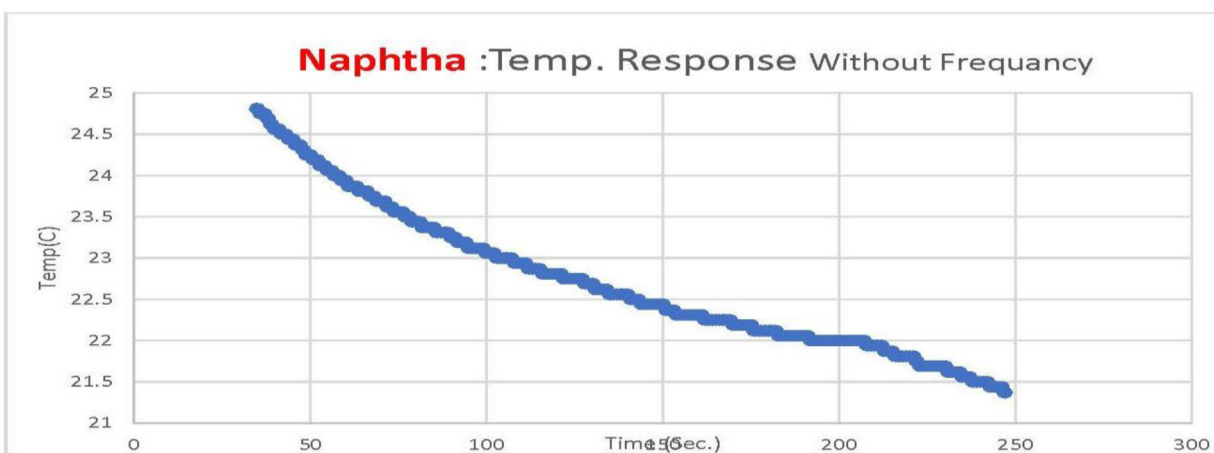
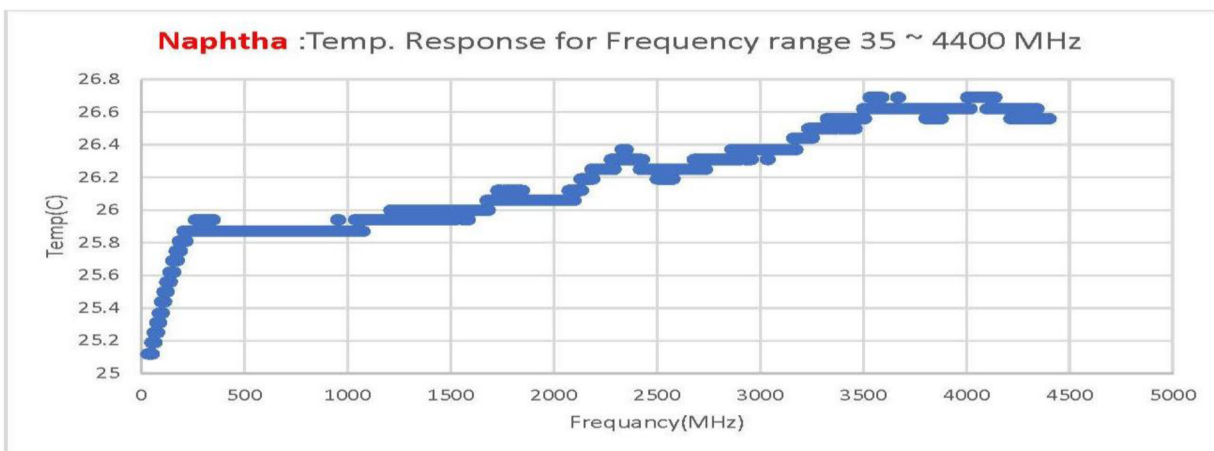


Fig. 1.7—Illustration of the Electromagnetic Pulse Generator and Temperature Monitoring Program

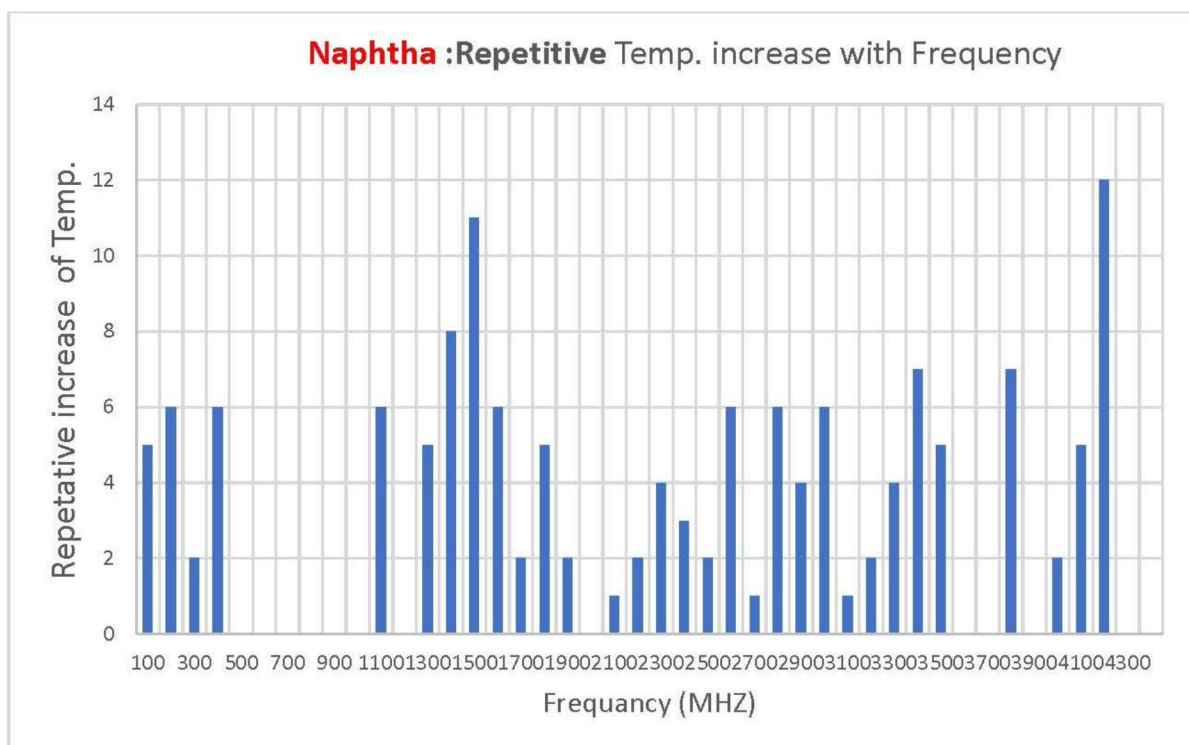
- **Data Collection and Curve Plotting:**
 - The temperature readings and corresponding frequencies were logged in an Excel sheet.
 - A curve was plotted to visualize the relationship between frequency and temperature, highlighting frequencies with significant temperature increases.



Graph 1:1—Relationship between Temperature Measurements and Radiation Frequencies from 35Mhz to 4400 Mhz for Naphtha

Naphtha Frequency Identification:

- **Analysis of Results:**
 - Frequencies with noticeable temperature increases were identified, indicating the frequencies where Naphtha absorbed electromagnetic waves most effectively.
- **Equations and Validation:**
 - Equations were formulated to analyze repeated temperature increases and their associated frequencies. Results were plotted in a new Bar chart to represent the frequency distribution of temperature increases across 100 MHz intervals.



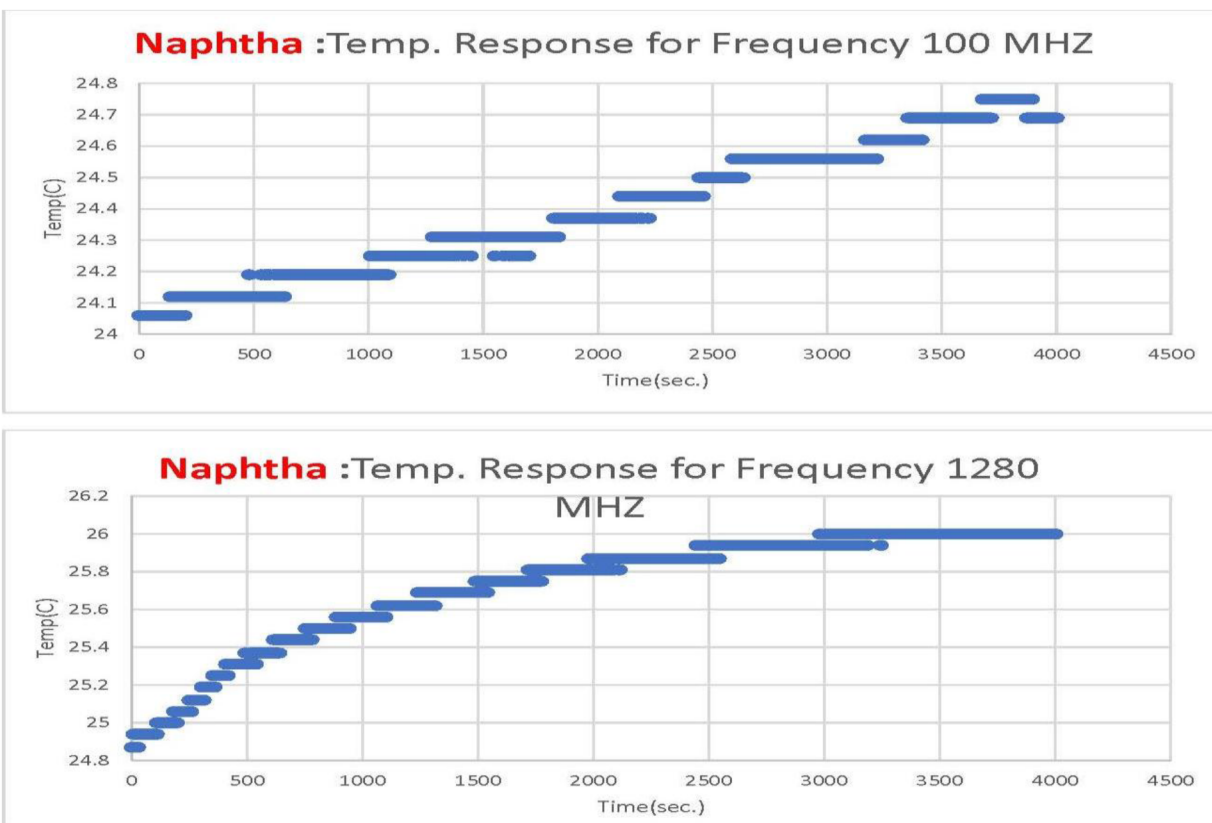
Graph 1.2—Relationship Between Frequency and Temperature Rise Repetition for Naphtha

- **Further Testing:**

- Naphtha was exposed to frequencies linked to these temperature increases for 4000 seconds, with temperature recorded every second.

- **Data Visualization:**

- Results were logged in an Excel sheet, and a curve was plotted to show the relationship between increase in temperature and time during the exposure with that frequency.



Graph 1.3—Relationship Between Temperature Measurements and Radiation Frequencies 100Mhz-1280Mhz for Naphtha (AS SAMPLE)

- **Result Compilation:**

- A "Table of Results" was created to display the relationship between frequency and temperature increase (ΔT), providing insights into frequencies with the highest electromagnetic wave absorption.

- **Naphtha Resonant Frequency identifying:**

- The resonant frequency was identified as the frequency with the highest absorption and maximum temperature increase during the 4000-second test period.

Table 1:2—Measurements of Temperature Differences Over Time to Determine Naphtha Resonance Frequency

Comparison of energy consumption for heating 1 cm ³ of Naphtha by 1C° at different frequencies								
	volume	14	cm ³					
	O/P power of pulse generator Board	3.2	mWatt /hr.					
No.	Acting frequency (MHZ)	MAX. haet Temp (C°)	MIN. haet Temp (C°)	Δ Temp. (C°)	Δ Time (sec)	Consumption power (mWatt)	Consumption power (mWatt) to heat 14 cm ³ by 1C°	Consumption power (mWatt) to heat 1 cm ³ by 1C°
1	100	24.75	24.06	0.69	4000	3.56	5.15	0.37
2	1280	26	24.87	1.13	4000	3.56	3.15	0.22
3	1450	30.44	28.75	1.69	4000	3.56	2.10	0.15
4	1510	26.19	25.37	0.82	4000	3.56	4.34	0.31
5	1735	25.5	24.87	0.63	4000	3.56	5.64	0.40
6	1900	25.94	25.12	0.82	4000	3.56	4.34	0.31
7	2200	26.25	25.19	1.06	4000	3.56	3.35	0.24
8	2952	25.75	25.25	0.5	3807	3.38	6.77	0.48
9	3130	25.62	25.31	0.31	4000	3.56	11.47	0.82
10	4000	24.3	24.12	0.18	2143	1.90	10.58	0.76
11	4200	24.62	23.87	0.75	4000	3.56	4.74	0.34
12	4400	25.81	25.25	0.56	4000	3.56	6.35	0.45

These resonance frequencies represent the specific frequencies at which each petroleum product demonstrates maximum absorption of electromagnetic waves, resulting in the most significant temperature increase. These findings offer crucial insights for optimizing the separation and refining of petroleum products using the proposed method.

As result of above: Naphtha resonance frequency is 1450 MHz

This process was repeated for other petroleum products, such as kerosene and diesel, as detailed in the referenced research papers.

Upon analyzing the data and results, the following conclusions were drawn:

The Research Result for Resonance Frequencies of Naphtha, Kerosene and Diesel fuel

- I. Naphtha resonance frequency: 1450 MHz
- II. Kerosene resonance frequency: 2000 MHz
- III. Diesel fuel resonance frequency: 2300 MHz

For a review of the detailed experiments study, refer to [Appendix C](#) Page No. (39).

By following these steps, the resonance frequency of the target product could be identifying and analyzed effectively.

Considerations in Scaling Up to an Industrial Model of Petroleum Refining Using the Innovative Method

1. Non-targeted petroleum components during resonance heating

- Regular cleaning procedures (steam, solvents, mechanical methods) will be applied.
- Proper design of draw-off systems will ensure that non-targeted components remain unaffected.

2. Effect of ambient offshore conditions (humidity, vibration)

- The system will be designed to resist offshore humidity and vibrations, ensuring no negative impact on accuracy.

3. Scaling from laboratory to offshore field application

- Full compliance with offshore safety standards (including explosion protection and operational hazards).
- Use of protective materials and enclosures to withstand harsh weather conditions.

4. Technical challenges

- No electromagnetic interference is expected, since each unit is dedicated to a specific petroleum product at its unique resonance frequency.
- Adequate cooling systems will be implemented to ensure proper heat dissipation.
- Resonance frequencies can be adjusted to accommodate variations in crude oil composition.

CONCLUSIONS

This research has introduced and validated a novel refining technique that employs electromagnetic radiation at resonance frequencies to selectively process crude oil fractions. The experimental work successfully identified resonance frequencies for Naphtha (1450 MHz), Kerosene (2000 MHz), and Diesel (2300 MHz), confirming the principle of selective molecular excitation. Compared to conventional distillation, the proposed method demonstrated a substantial reduction in energy consumption—up to 45%—and a corresponding decrease in carbon emissions ranging between 18% and 40%, depending on operating conditions.

The findings highlight not only the efficiency and environmental advantages of resonance-based refining but also its unique suitability for offshore and modular applications, where footprint, safety, and energy optimization are critical. Unlike large distillation towers, compact magnetron units can provide targeted fractionation with minimal infrastructure requirements.

Although challenges remain in scaling the technology, including industrial frequency control, long-term thermal management, and adaptation to varying crude compositions, the results establish strong proof-of-concept evidence. The proposed method represents a promising step toward sustainable, energy-efficient petroleum refining. Future research should focus on pilot-scale validation, economic feasibility, and integration with existing refinery operations to accelerate industrial adoption of this innovative approach.

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Appendix A

Detail study for Economic and Oil & Gas Advantages of Fully Implementing the Research Idea

1. Savings in heating power by 45%.

Energy Savings in Crude Oil Refining Using Electromagnetic Waves Instead of Distillation Towers
– A Case Study on One Barrel can save approximately 45 % of distillation heat energy.

Introduction

Traditional crude oil refining relies on heating the entire barrel to 350°C inside distillation towers, consuming substantial energy.

In contrast, the proposed method uses electromagnetic waves at the resonance frequencies of petroleum products (naphtha, kerosene, diesel), allowing selective heating and separation, drastically reducing energy consumption.

This approach takes advantage of the physical concept that the specific heat capacity of crude oil is 2.5 (kJ/kg°C), while specific heat capacities of naphtha, kerosene, and diesel fuel are 2.0, 2.1, and 2.2 (kJ/kg °C) respectively.

- Heating from 25°C up to:
 - Naphtha: 200°C
 - Kerosene: 275°C
 - Diesel: 350°C
 - Crude Oil: 350°C
- Densities used (approximate):
 - Naphtha: 0.75 kg/L
 - Kerosene: 0.80 kg/L
 - Diesel: 0.83 kg/L
 - Crude oil: 0.85 kg/L
- Now we can calculate the heat energy:
- $Q = m * c * \Delta T$
- Where: Q = heat energy (in joules) m = mass of naphtha (in grams) c = specific heat capacity of naphtha (in J/g°C) ΔT = temperature change (final temperature - initial temperature).

Product	Mass (kg)	Densities	Volume	Specific Heat (kJ/kg·K)	ΔT (°C)	Conventional Heating (kJ)	EM Heating (kJ)	Energy Saved (kJ)	Saving (%)
Naphtha	30	0.75	40	2	175		10,500	62,019	56.5%
Kerosene	22	0.8	28	2.1	250		11,550		
Diesel	36	0.83	43	2.2	325		25,740		
Crude Oil	135.15	0.85	159	2.5	325	109,809			
Total Energy						109,809	47,790		

The percentage of power savings achieved by selectively heating specific petroleum products within crude oil is 56.5%.

By assume that the efficiency of magnetron is 80%. So that the final

The percentage of power savings achieved by selectively heating specific petroleum products within crude oil is 45%.

2. Reduction in Carbon Emissions During the Refining Process: From 18% Up to 40%.

The distillation tower accounts for approximately 40% of a refinery's total energy consumption. The proposed method reduces the heat energy required to refine crude oil by up to 45%, resulting in an 18% reduction in carbon emissions. Furthermore, using renewable energy sources to power the electromagnetic refining units can lead to a total reduction in carbon emissions of up to 40%.

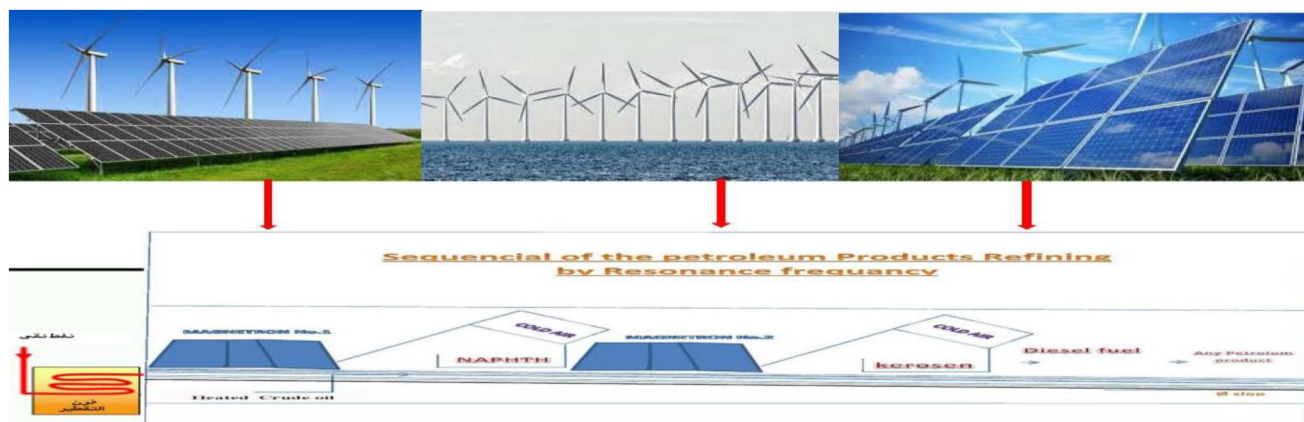


Fig. 1.10—Illustration of Renewable Energy Sources Used in Electromagnetic Wave Refining

Overall, the implementation of this new method offers cost savings, flexibility in capacity expansion, streamlined operations, enhanced workplace safety, and the potential for offshore refining stations. These economic benefits contribute to the efficiency and profitability of the crude oil refining process.

3. Economic Benefits of the Offshore Refining Stations:

The establishment of offshore refining stations near extraction platforms offers notable logistical advantages. This approach brings about several benefits, including:

- I. **Direct Loading of Tankers:** By situating the refining stations offshore, petroleum products can be loaded directly onto tankers from the facilities. This eliminates the need for pipelines and associated infrastructure, streamlining the transportation process and reducing costs.
- II. **Infrastructure Cost Reduction:** Offshore refining stations eliminate the requirement for extensive infrastructure, such as pipelines and storage facilities. This reduction in infrastructure needs translates to cost savings and improved operational efficiency.
- III. **Efficient Transportation:** Direct loading of tankers at offshore refining stations ensures a more efficient transportation process. Petroleum products can be shipped directly from the production site, reducing the distance and time required for transportation. This results in quicker delivery to markets and enhanced supply chain efficiency.
- IV. **Flexibility and Scalability:** Offshore refining stations offer flexibility in terms of scalability and expansion. New facilities can be constructed or existing ones modified to accommodate changing production volumes or the addition of new extraction platforms. This adaptability supports the efficient utilization of resources and allows for adjustments in line with market demands.
- V. **Environmental Considerations:** Offshore refining stations can contribute to environmental protection by minimizing infrastructure and potential disruptions to sensitive ecosystems. The offshore location reduces the risk of spills or accidents near coastal areas, preserving the environment and minimizing environmental impact.

By establishing offshore refining stations alongside extraction platforms, companies can benefit from direct loading of tankers, reduced infrastructure costs, improved transportation efficiency, scalability, and environmental considerations. These advantages contribute to cost savings, operational efficiency, and sustainable practices in the petroleum industry.



Fig. 1.9—Illustration of Applicable of Offshore Refining Platforms by using the Research Refinery Unit and Renewable Energy from Wind Tower.

4. Economic Benefits of the Quick Production and Assembly:

The implementation of module systems in refining units enables rapid production and assembly, resulting in significant time and resource savings. The following points highlight the benefits of this streamlined process:

- I. **Efficient Manufacturing:** Module systems involve the prefabrication of standardized components that can be produced in controlled manufacturing environments. This approach allows for efficient production with optimized workflows, minimizing production time and costs.
- II. **Simplified Assembly:** The modular nature of the refining units simplifies the assembly process. The standardized components can be easily transported to the site and seamlessly interconnected, reducing the complexity and duration of the assembly phase.
- III. **Time-Saving:** The use of module systems significantly reduces the overall project timeline. The concurrent production and assembly of modules enable parallel workflows, eliminating the need for sequential construction processes. As a result, refining operations can commence sooner, reducing downtime and accelerating time-to-market.
- IV. **Resource Optimization:** Module systems enable better resource management by minimizing waste and ensuring efficient utilization of materials. The controlled manufacturing environment allows for precise production planning and reduced material wastage, resulting in cost savings and environmental benefits.
- V. **Scalability and Flexibility:** The modular design of the refining units offers scalability and flexibility. Additional modules can be easily integrated into the existing setup, allowing for future expansion or modifications to accommodate changing production requirements.
- VI. **Quality Control:** The standardized production process in module systems allows for stringent quality control measures. Each module can undergo thorough testing and inspection before integration, ensuring high-quality construction and reliable performance.

By employing module systems, companies can benefit from quick production and assembly of refining units, saving time, reducing costs, and improving operational efficiency. The streamlined process enables faster project completion, enhanced resource optimization, and the flexibility to adapt to evolving industry demands.

5. Saving in construction costs

The new method of extracting petroleum products through horizontal units specialized in each specific product offers substantial cost savings compared to traditional distillation towers. Here are the key factors contributing to the cost-effectiveness of the new method:

- I. **Construction Costs:** Traditional distillation towers involve significant construction costs due to their large size and complex structures. In contrast, the electromagnetic wave horizontal extraction units are smaller and simpler in design, resulting in lower construction costs.
- II. **Weight Reduction:** Distillation towers typically have considerable weight due to the presence of crude oil, petroleum products, and the heavy metal structure of the tower. This weight can impact the costs associated with civil foundations for the tower. In contrast, the horizontal units used in the new method are specialized for each petroleum product, resulting in reduced weight and less strain on the foundation, leading to cost savings.
- III. **Simplified Design:** The design of the electromagnetic wave horizontal extraction units is more straightforward compared to traditional distillation towers. The focused nature of these units allows for optimized construction and reduced complexity, resulting in overall cost savings.

By implementing the new method of extracting petroleum products through specialized horizontal units, companies can achieve significant cost savings in construction expenses, reduced weight impact on foundations, and simplified designs. These cost benefits make the new method a compelling option for the refining industry, promoting efficiency and economic viability.



Fig. 1.8—Illustration of Electromagnetic Wave Refinery Unit with respect to distillation towers

6. Economic benefited of the capacity Expansion Flexibility:

The horizontal configuration of the refining units offers a distinct advantage in terms of flexibility for capacity expansion. This flexibility allows for seamless adjustments to accommodate changing market demands and the need to increase production volumes within constrained timelines and cost limitations. The following points highlight the benefits of this flexibility:

- I. **Easy Scalability:** The horizontal design of the refining units enables straightforward scalability. Additional units can be seamlessly incorporated into the existing setup, allowing for increased production capacity as required. This ease of scalability ensures quick response times to market demands without significant disruptions to operations.
- II. **Rapid Deployment:** The modular nature of the horizontal units facilitates rapid deployment of additional capacity. New units can be efficiently installed and integrated into the system, minimizing downtime, and maximizing production output in a shorter timeframe.
- III. **Cost-Effective Expansion:** The flexibility in capacity expansion provided by the horizontal configuration allows for cost-effective growth. Instead of building entirely new facilities or investing in extensive modifications to existing structures, additional units can be added incrementally, reducing capital expenditure and optimizing cost-efficiency.

- IV. **Seamless Integration:** The horizontal units can be seamlessly integrated into the existing production line. This integration ensures smooth operation and compatibility with the current processes, minimizing disruptions and maintaining overall operational efficiency.
- V. **Adaptability to Market Demands:** The capacity expansion flexibility offered by the horizontal units allows refineries to quickly adapt to fluctuating market demands. Whether there is a surge in product demand or a need to adjust production volumes, the system can readily accommodate these changes, ensuring efficient utilization of resources and timely response to market dynamics.

In summary, the horizontal configuration of the refining units provides capacity expansion flexibility, enabling refineries to respond swiftly and cost-effectively to changing market demands. The ability to scale up production capacity in a timely manner and without significant disruptions contributes to enhanced operational efficiency and competitiveness in the industry.

7. Economic Benefits of the Compact Refining Facilities:

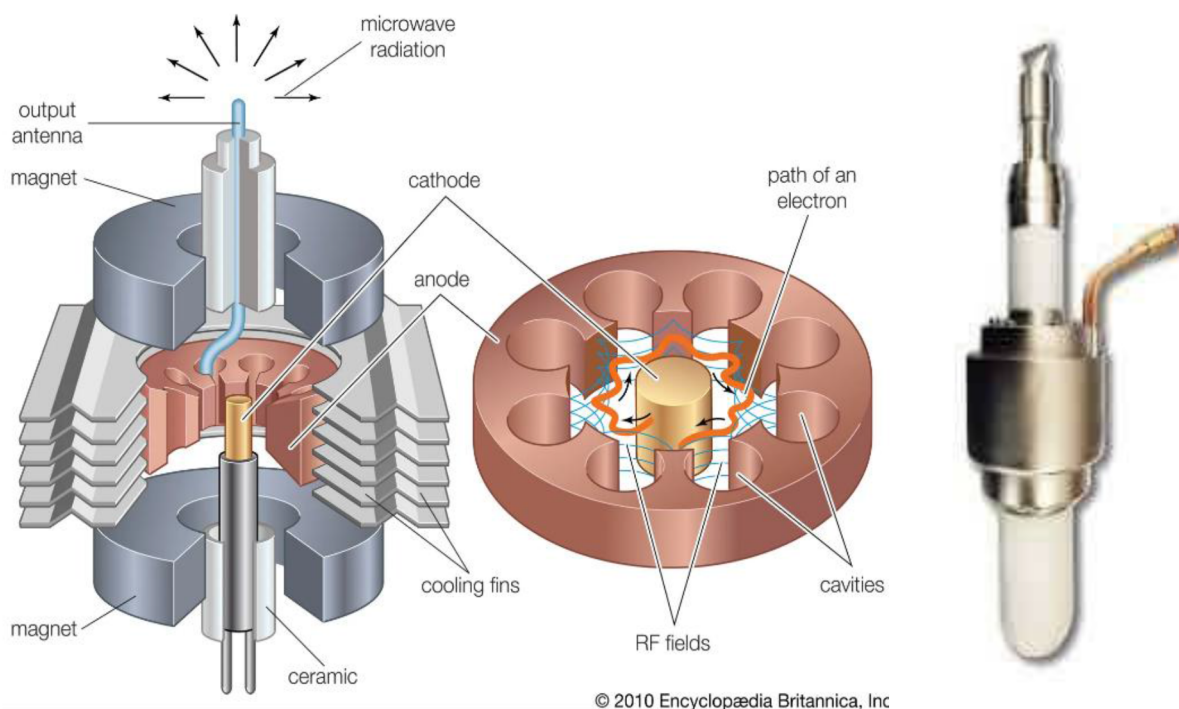
The capability to establish compact refining facilities presents notable advantages, particularly in regions with limited space, such as smaller geographic areas and developing countries. The following benefits highlight the value of compact facilities in such contexts:

- I. **Local Market Focus:** Compact refining facilities can cater specifically to local markets, effectively meeting the demand for petroleum products within the immediate vicinity. By eliminating the need for extensive transportation and distribution networks, these facilities reduce associated costs and logistical complexities.
- II. **Cost Efficiency:** With the proximity of compact facilities to their target markets, transportation costs are significantly minimized. The reduced need for long-distance shipping or trucking of petroleum products translates into lower expenses, resulting in cost-efficient operations.
- III. **Enhanced Service Provision:** By situating refining facilities closer to local markets, the availability and accessibility of petroleum products can be improved. This enhanced service provision benefits communities, businesses, and industries reliant on a stable supply of refined products.
- IV. **Reduced Storage Requirements:** Compact facilities typically produce petroleum products in quantities that align with the local demand. This avoids the need for extensive storage capacities and associated costs, making efficient use of available space.
- V. **Economic Development:** The establishment of compact refining facilities in developing countries can contribute to local economic growth. These facilities generate employment opportunities and foster a self-sufficient and sustainable petroleum product supply chain within the region.

In summary, compact refining facilities offer advantages in terms of localized market focus, cost efficiency, improved service provision, reduced storage requirements, and economic development. Their suitability for smaller geographic areas and developing countries makes them a practical solution for meeting local demand, reducing transportation costs, and enhancing overall operational efficiency.

Appendix B

Magnetron instruction and efficiency:



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Cavity magnetron

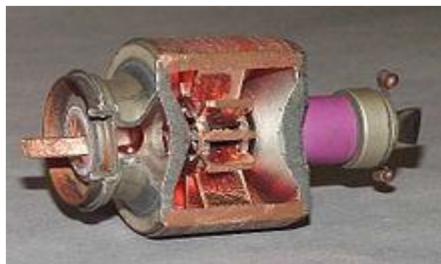
Fig. 1.3—Illustration of the Structural Parts of a Magnetron

Cavity magnetron

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"Magnetron" redirects here. It is not to be confused with [Megatron](#), [Metatron](#), or [Magneton](#) (disambiguation).



Magnetron with section removed to exhibit the cavities. The cathode in the center is not visible. The antenna emitting microwaves is at the left. The magnets producing a field parallel to the long axis of the device are not



A similar magnetron with a different section removed. Central cathode is visible; antenna conducting microwaves at the top; magnets are not shown.



Obsolete 9 GHz magnetron tube and magnets from a Soviet aircraft radar. The tube is embraced between the poles of two horseshoe-shaped alnico magnets (*top, bottom*), which create a magnetic field along the axis of the tube. The microwaves are emitted from the waveguide aperture (*top*) which in use is attached to a waveguide conducting the microwaves to the radar antenna. Modern tubes use rare-earth magnets, electromagnets or ferrite magnets which are much less bulky.

The **cavity magnetron** is a high-power vacuum tube that generates microwaves using the interaction of a stream of electrons with a magnetic field while moving past a series of cavity resonators which are small, open cavities in a metal block. Electrons pass by the cavities and cause microwaves to oscillate within, similar to the functioning of a whistle producing a tone when excited by an air stream blown past its opening. The resonant frequency of the arrangement is determined by the cavities' physical dimensions. Unlike other vacuum tubes, such as a klystron or a traveling-wave tube (TWT), the magnetron cannot function as an amplifier for increasing the intensity of an applied microwave signal; the magnetron serves solely as an oscillator, generating a microwave signal from direct current electricity supplied to the vacuum tube.

Table 1.1—TYPE OF MAGNETRON WITH RELEVANT CHARACTERISTIC

Anode	Duty	Frequency (GHz)	Output power (kW)	Anode current (A)	Impedance (k Ω)	Overall Efficiency (%)
Strapped	CW	0.9	30–100	3–6	3.0–5.0	80–90
Strapped	CW	2.45	0.2–30	2–40	3.0–7.0	65–70
Strapped	pulse	2–4	5–4500	4–130	0.5–1.5	35–50
Strapped	pulse	4–8	200–1000	25–80	0.5–1.0	30–50
Strapped	pulse	8–11	1–2000	2–100	0.4–1.4	30–50
Strapped	pulse	11–18	20–250	8–28	0.8–1.2	20–45
Rising sun	pulse	25–35	40–125	13–28	0.7–1.0	20–25
Coaxial	pulse	8–10	10–350	4–27	0.8–1.1	30–50
Coaxial	pulse	14–17	1–180	2–25	0.8–1.9	20–40
Coaxial	pulse	32–36	25–130	10–20	1.0–1.4	20–30
Long anode	pulse	3	2000–5500	50–200	0.2–0.5	45–55

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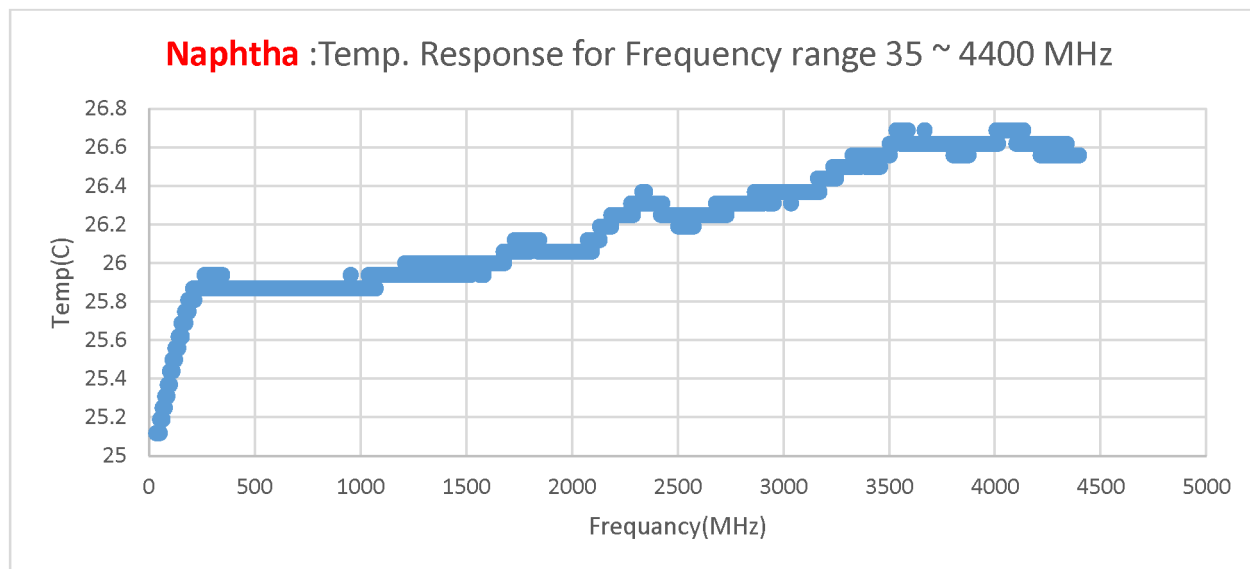
Magnetrons (Chapter 15) - Microwave and RF Vacuum Electronic Power Sources

Appendix C

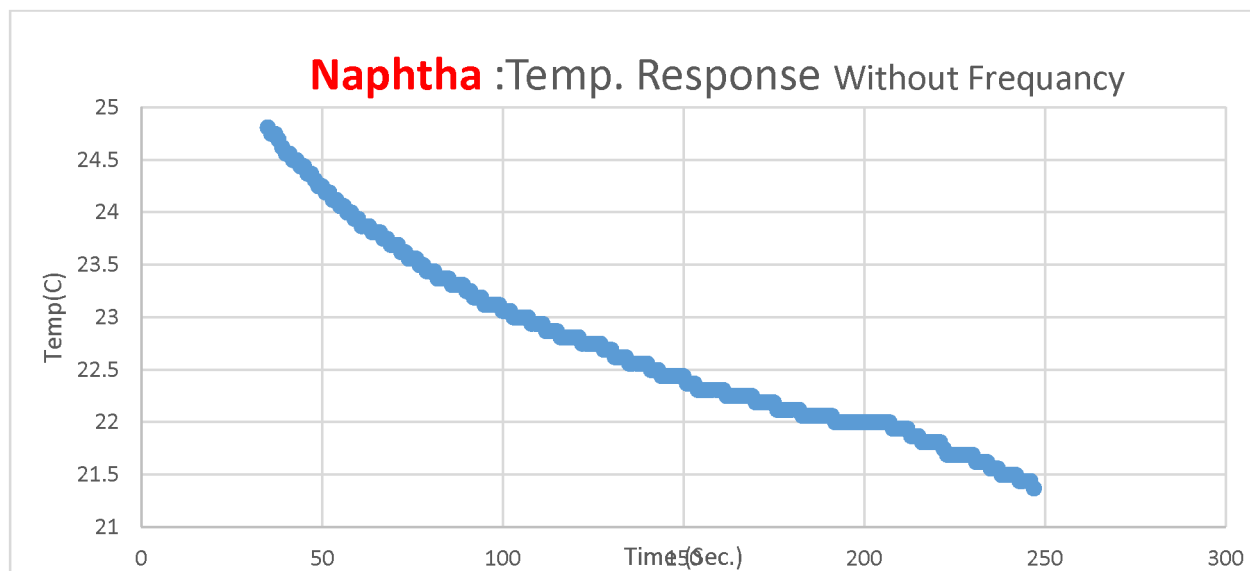
Experimental study and results:

Experimental Study for Identify Naphtha Resonance Frequency.

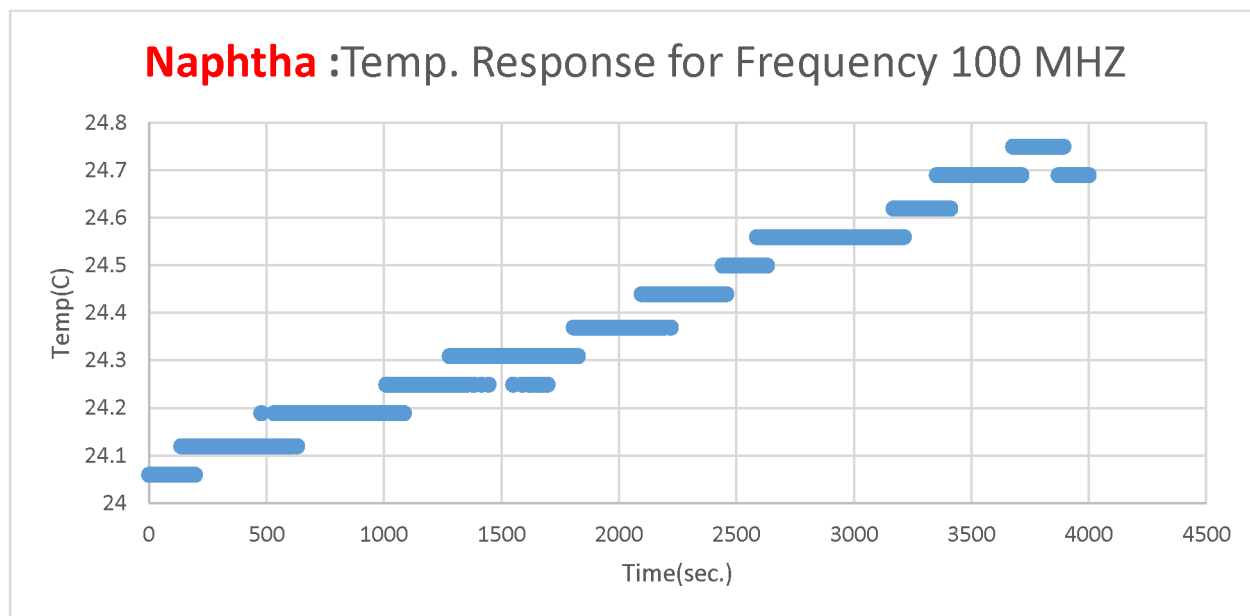
Chart The Relation Between Radiated Frequency and The Recorded Temperature for Naphtha.



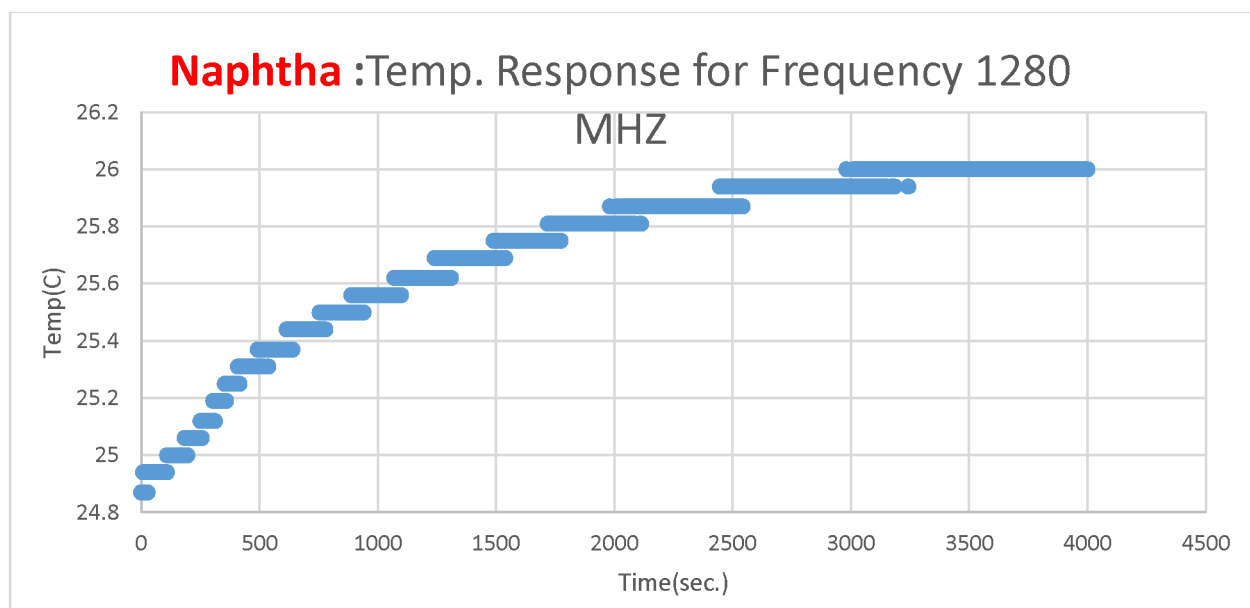
Graph 1.4—Relationship Between Temperature Measurements and Radiation Frequencies from 35Mhz to 4400 Mhz for Naphtha



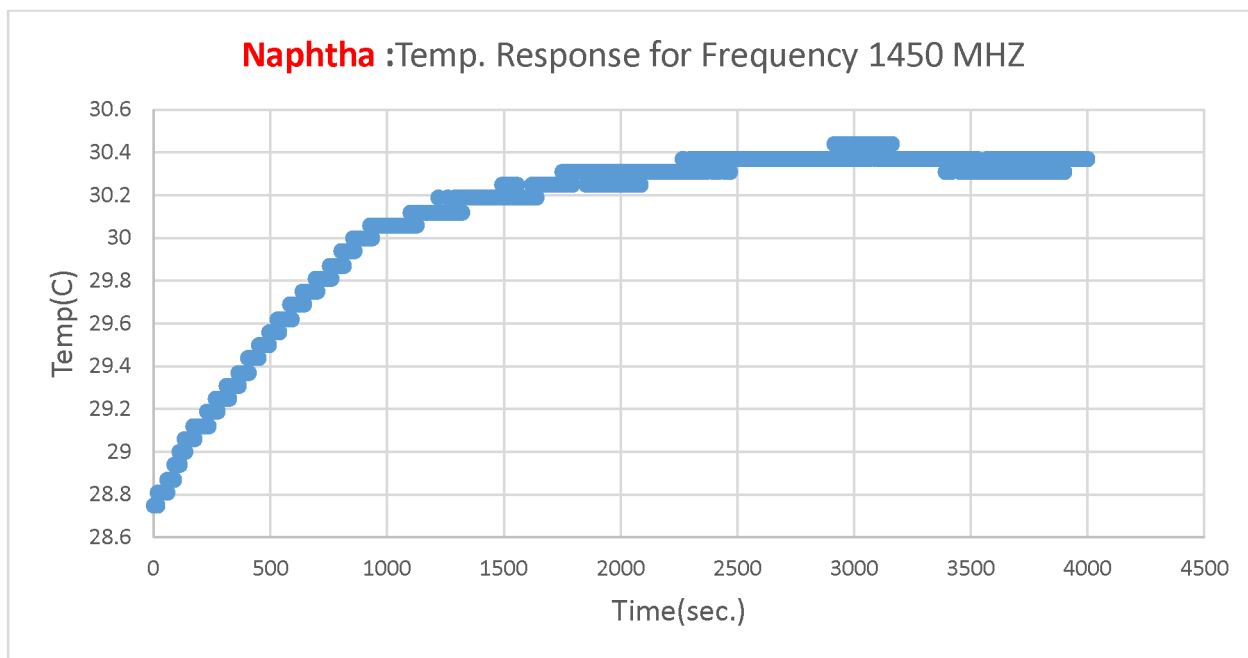
Graph 1.5—Relationship Between Temperature Measurements and Time without radiation for Naphtha



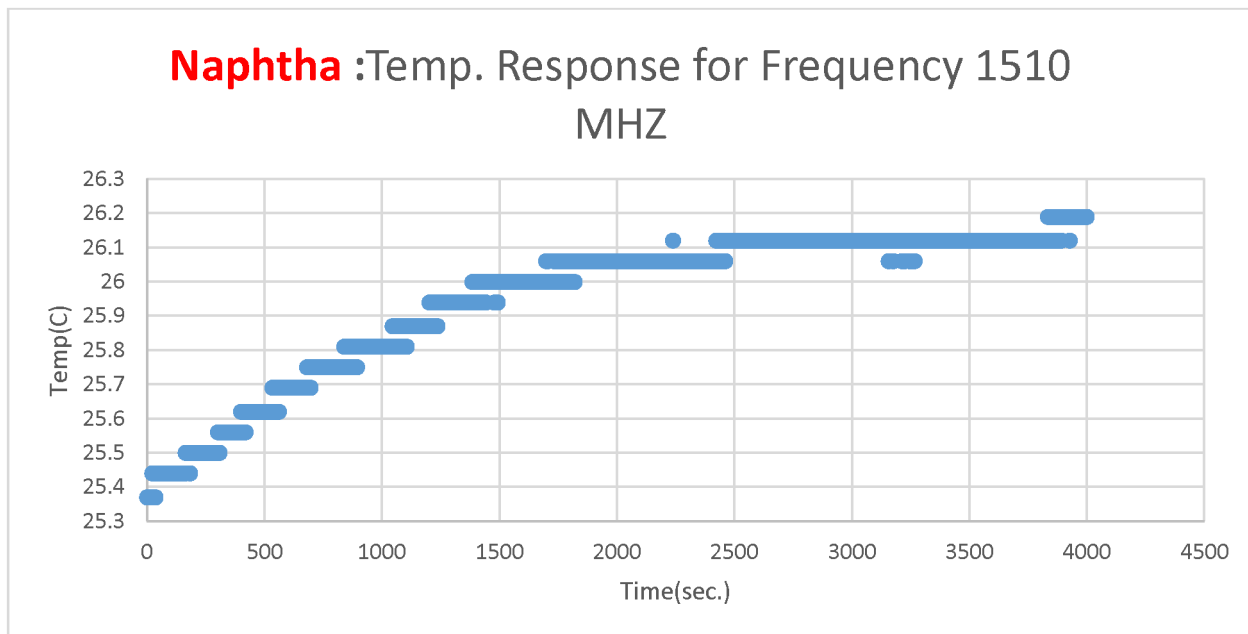
Graph 1.6—Relationship Between Temperature Measurements and Radiation Frequency at 100Mhz for Naphtha



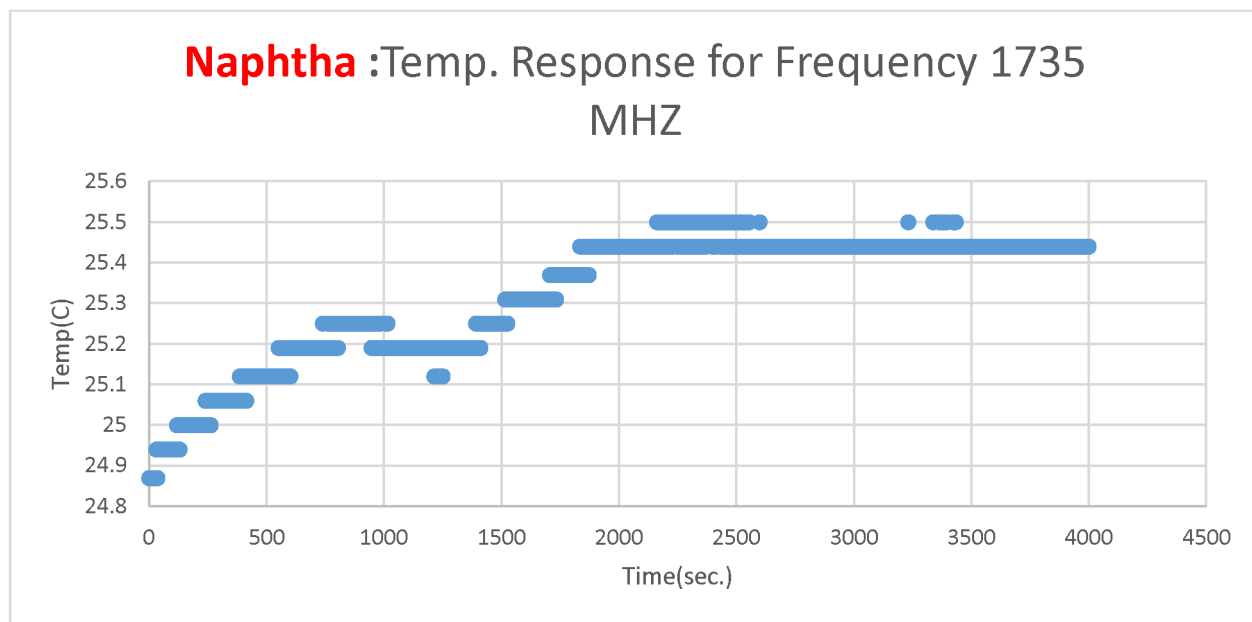
Graph 1.7—Relationship Between Temperature Measurements and Radiation Frequency at 1280 Mhz for Naphtha



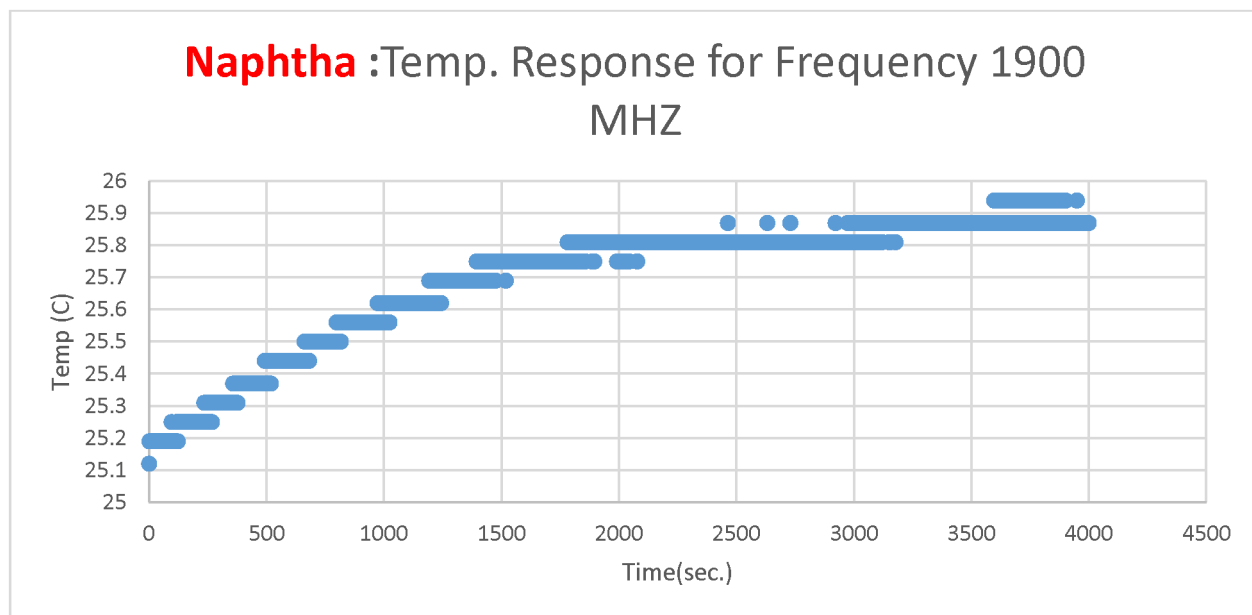
Graph 1.8—Relationship Between Temperature Measurements and Radiation Frequency at 1450 Mhz for Naphtha



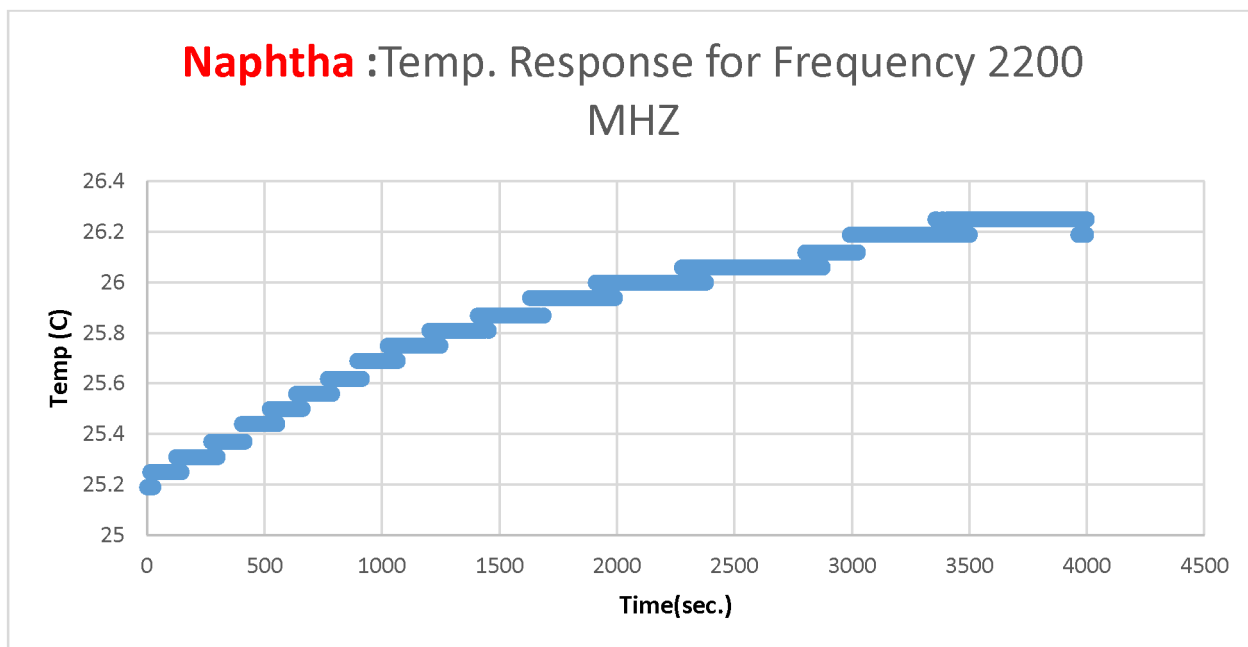
Graph 1.9—Relationship Between Temperature Measurements and Radiation Frequency at 1510 Mhz for Naphtha



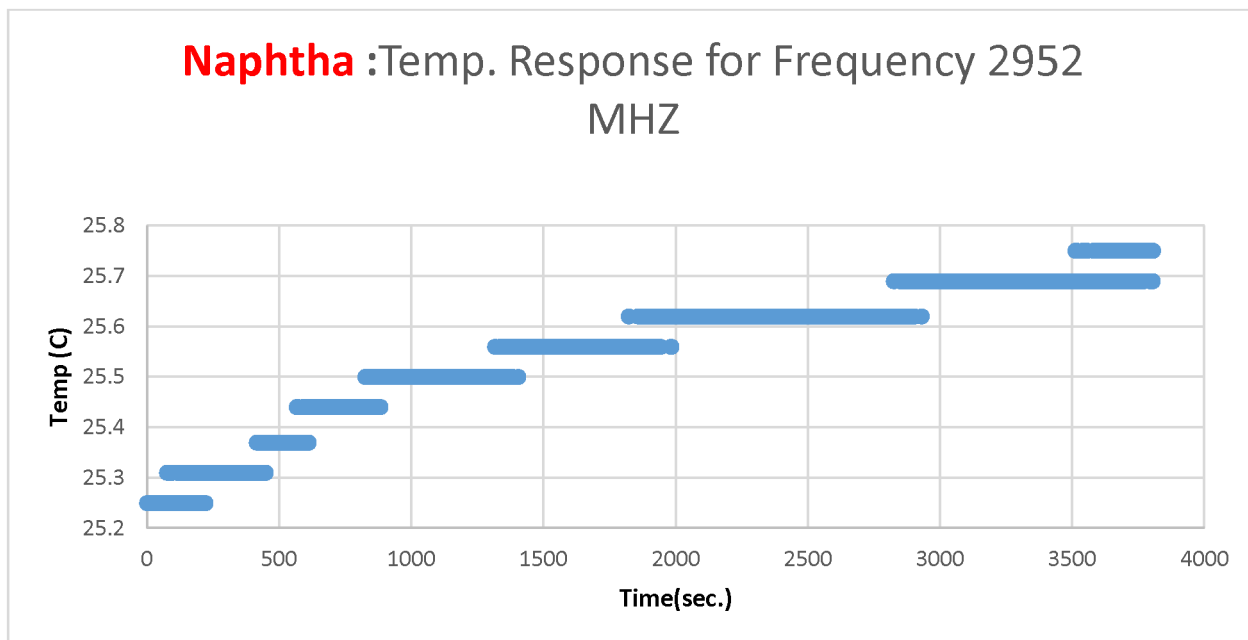
Graph 1.10—Relationship Between Temperature Measurements and Radiation Frequency at 1735 Mhz for Naphtha



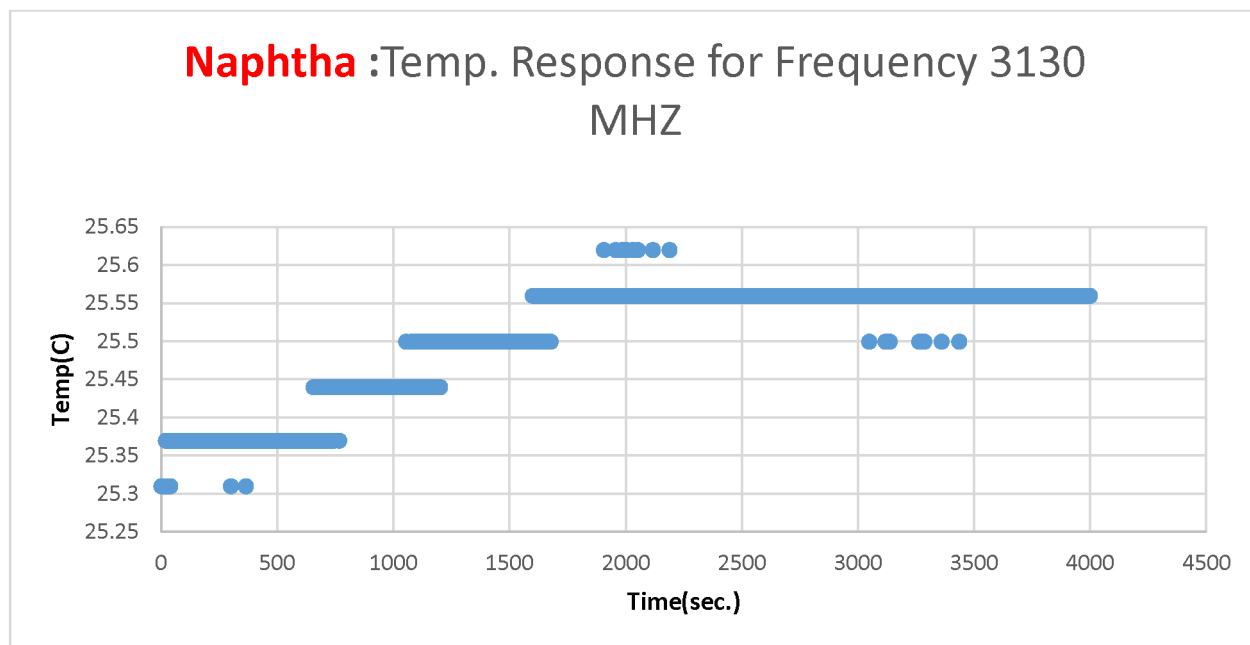
Graph 1.11—Relationship Between Temperature Measurements and Radiation Frequency at 1900 Mhz for Naphtha



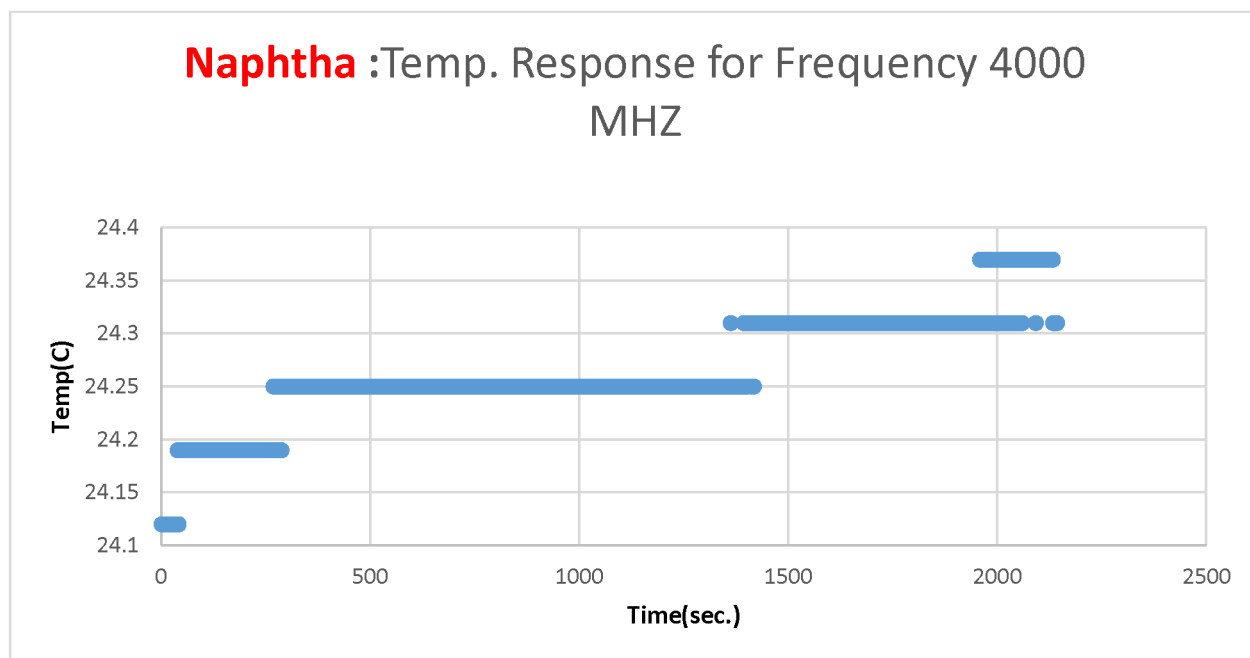
Graph 1.12—Relationship Between Temperature Measurements and Radiation Frequency at 2200 Mhz for Naphtha



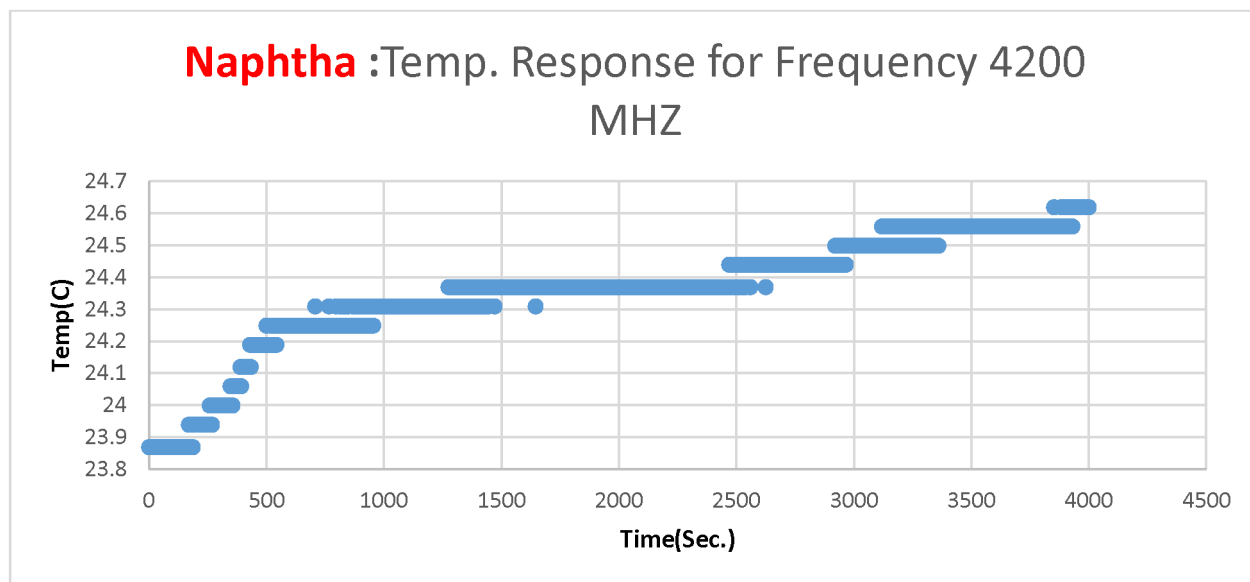
Graph 1.13—Relationship Between Temperature Measurements and Radiation Frequency at 2952 Mhz for Naphtha



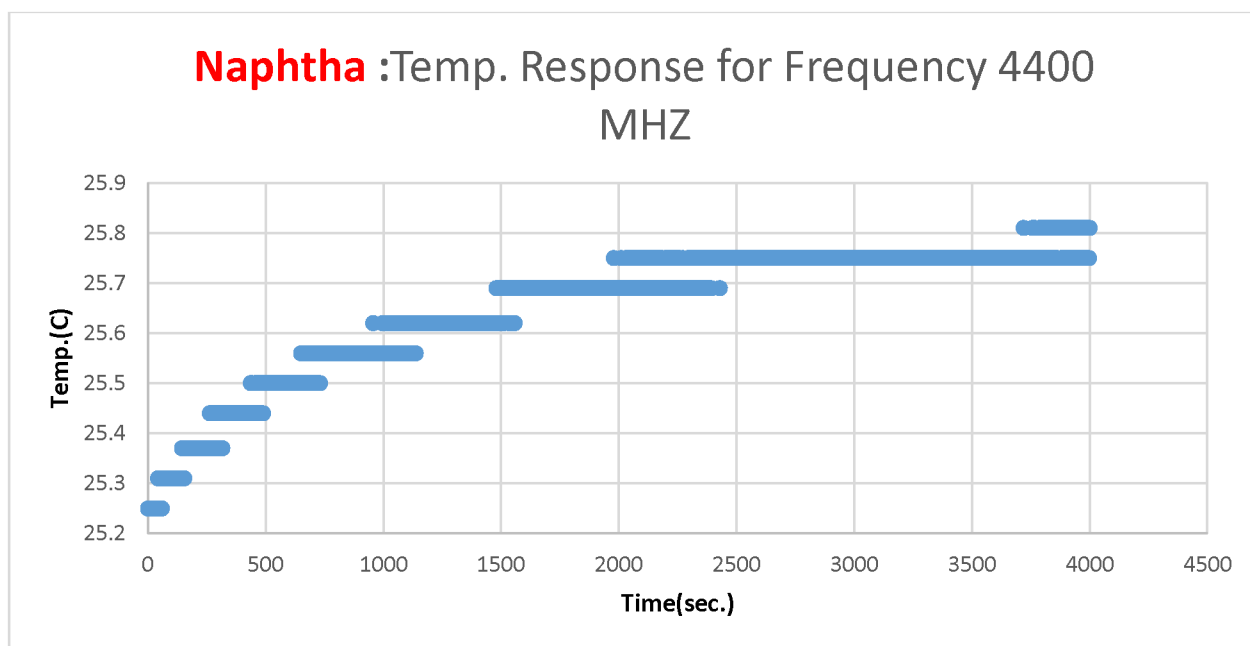
Graph 1.14—Relationship Between Temperature Measurements and Radiation Frequency at 3130 Mhz for Naphtha



Graph 1.15—Relationship Between Temperature Measurements and Radiation Frequency at 4000 Mhz for Naphtha

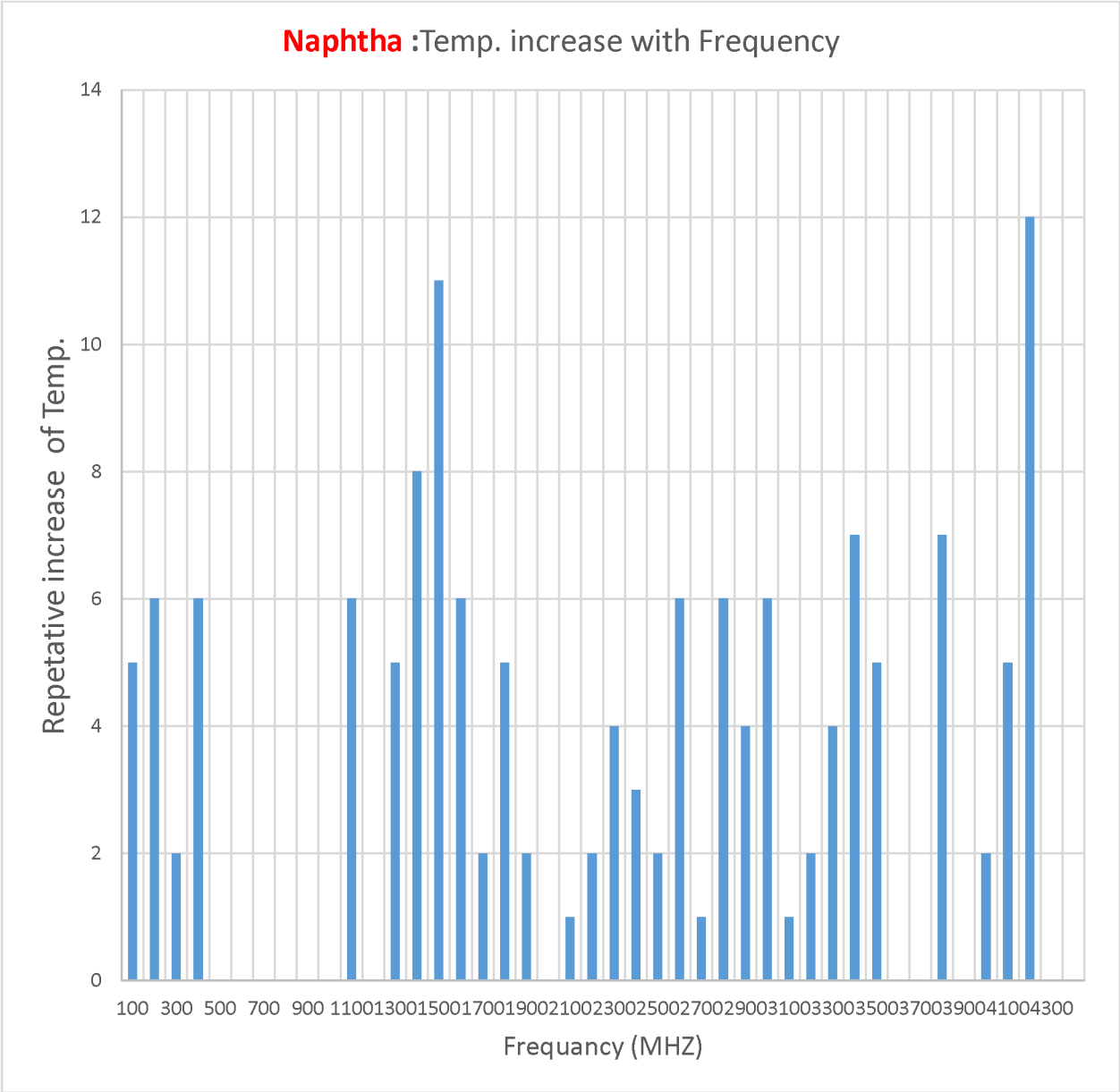


Graph 1.16—Relationship Between Temperature Measurements and Radiation Frequency at 4200 Mhz for Naphtha



Graph 1.17—Relationship Between Temperature Measurements and Radiation Frequency at 4400 Mhz for Naphtha

Repeated temperature increases and their associated frequencies Chart for Naphtha.



Graph 1.18—Relationship Between Frequency and Temperature Rise Repetition for Naphtha.

Table 1.3—Measurements of Temperature Differences Over Time to Determine Naphtha Resonance Frequency

**Naphtha Comparison of energy consumption Table
for heating 1 cm³ of Naphtha by 1C° at different frequencies**

volume :14 cm³

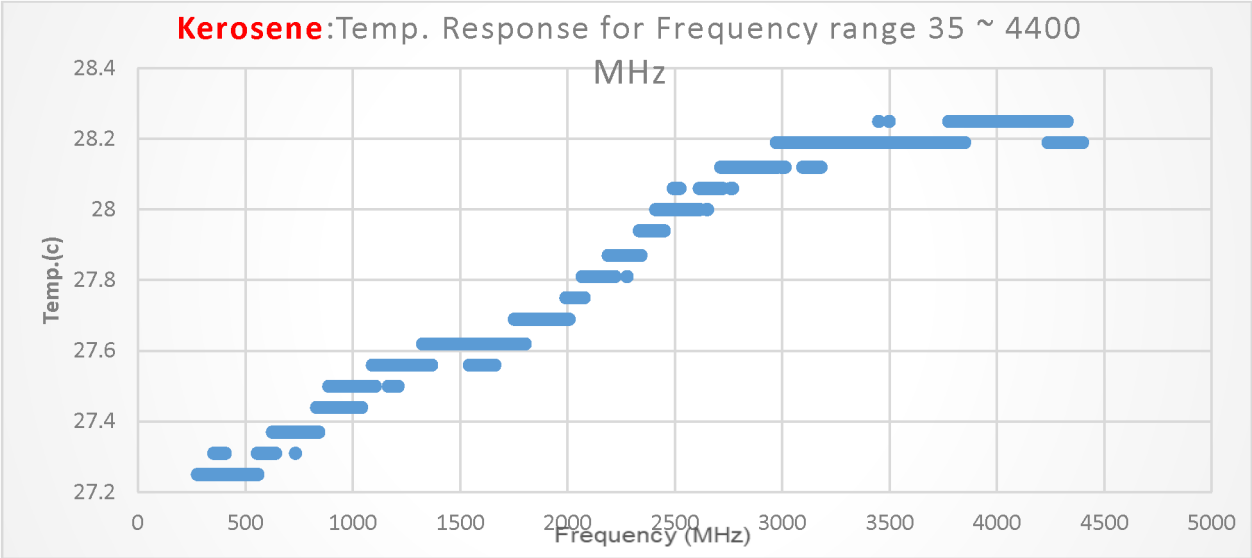
O/P power of pulse generator Board: 3.2 mWatt /hr.

No .	Acting frequency (MHZ)	MAX. heat Temp (C°)	MIN. heat Temp (C°)	Δ Temp . (C°)	Δ Time (sec)	Consumption power (mWatt)	Consumption power (mWatt) to heat 14 cm ³ by 1C°	Consumption power (mWatt) to heat 1 cm ³ by 1C°
1	100	24.75	24.06	0.69	4000	3.56	5.15	0.37
2	1280	26	24.87	1.13	4000	3.56	3.15	0.22
3	1450	30.44	28.75	1.69	4000	3.56	2.10	0.15
4	1510	26.19	25.37	0.82	4000	3.56	4.34	0.31
5	1735	25.5	24.87	0.63	4000	3.56	5.64	0.40
6	1900	25.94	25.12	0.82	4000	15.56	18.97	1.36
7	2200	26.25	25.19	1.06	4000	3.56	3.35	0.24
8	2952	25.75	25.25	0.5	3807	3.38	6.77	0.48
9	3130	25.62	25.31	0.31	4000	3.56	11.47	0.82
10	4000	24.3	24.12	0.18	4000	3.56	19.75	1.41
11	4200	24.62	23.87	0.75	4000	3.56	4.74	0.34
12	4400	25.81	25.25	0.56	4000	3.56	6.35	0.45

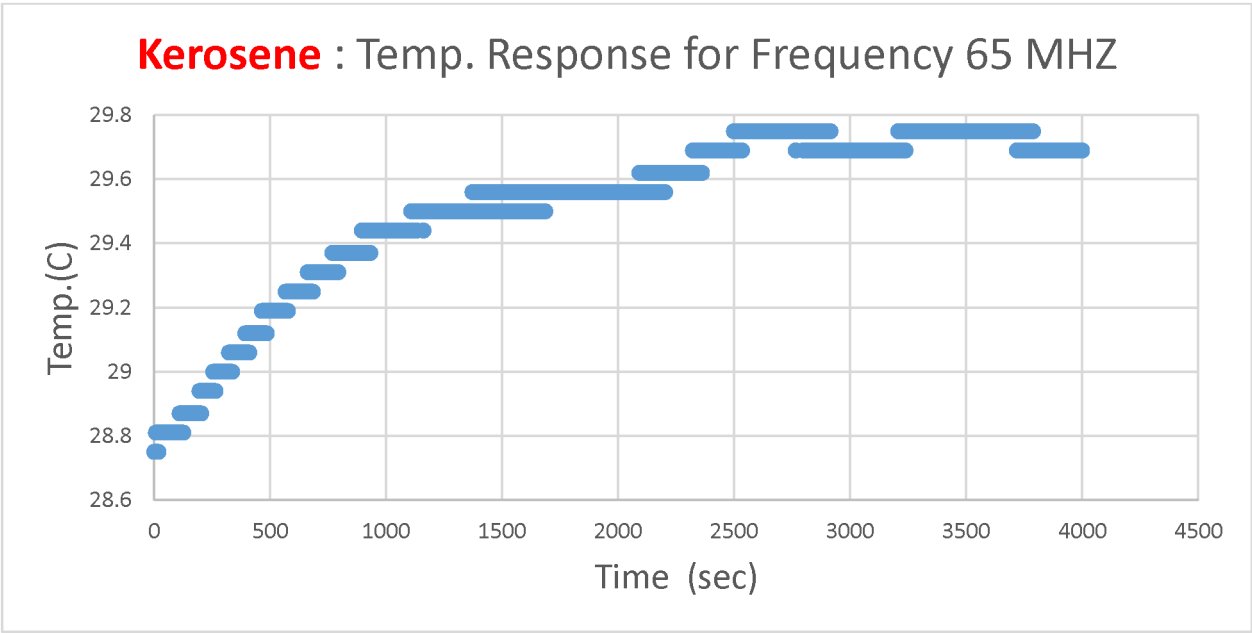
As result of above: Naphtha resonance frequency is 1450 MHz

Experimental Study for Identify Kerosene Resonance Frequency.

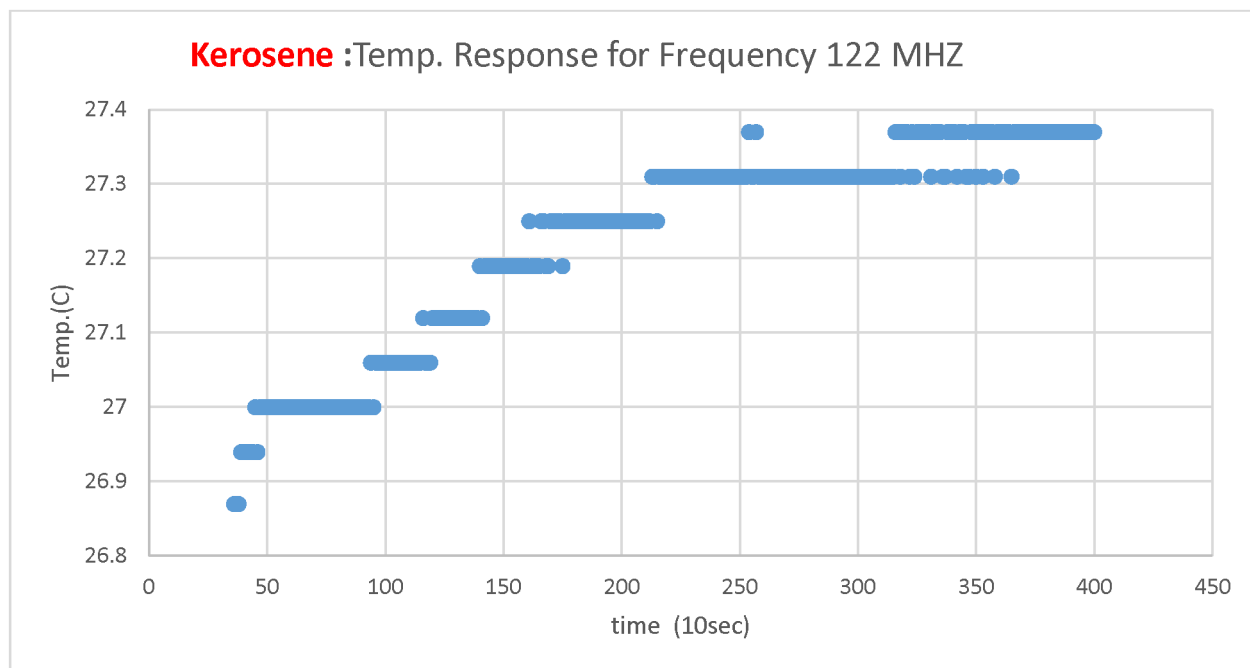
Chart The Relation Between Radiated Frequency and The Recorded Temperature for Kerosene.



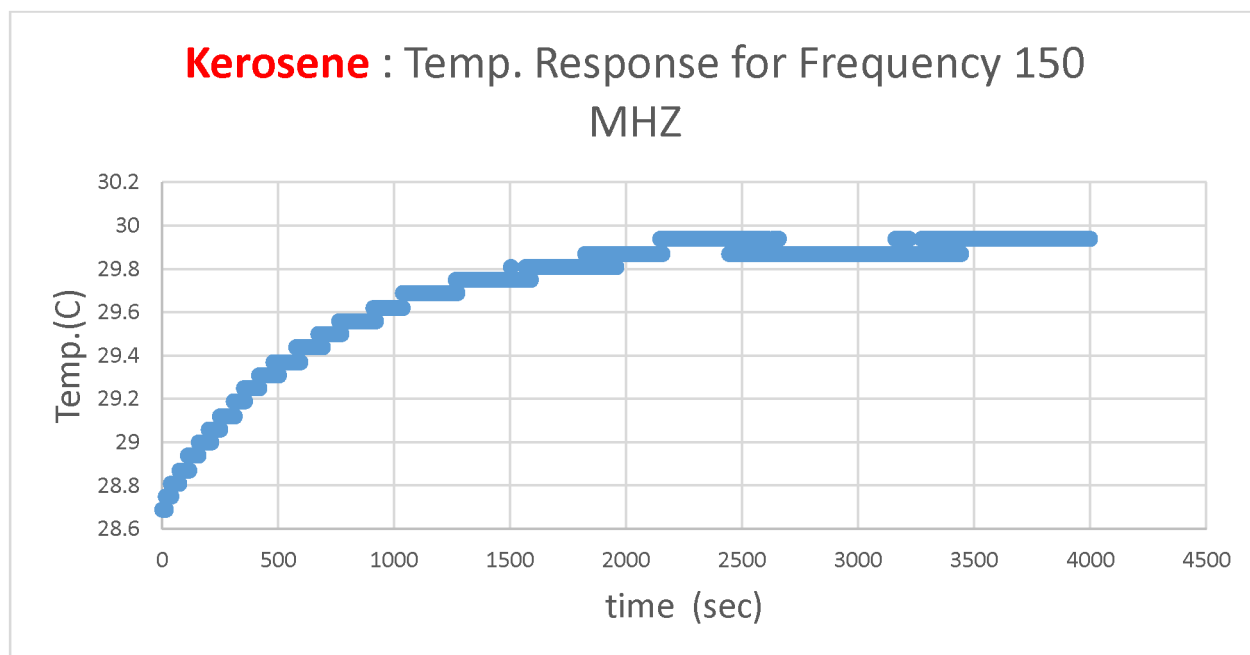
Graph 1.19—Relationship Between Temperature Measurements and Radiation Frequencies from 35 Mhz to 4400 Mhz for Kerosene



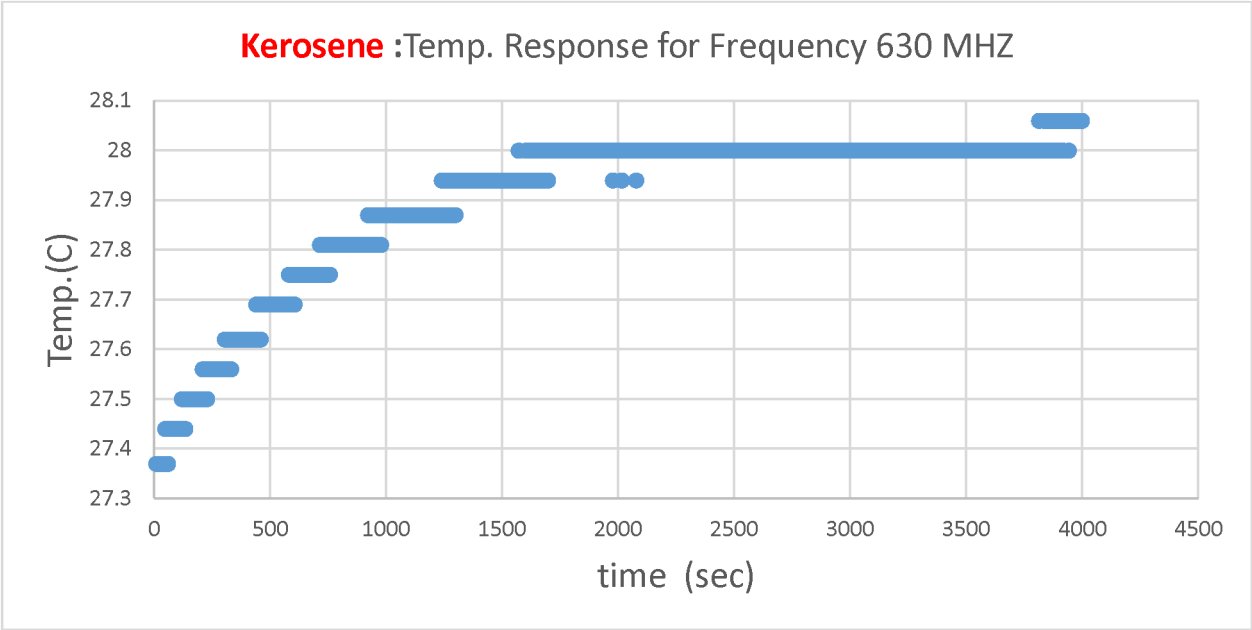
Graph 1.20—Relationship Between Temperature Measurements and Radiation Frequency at 65 Mhz for Kerosene



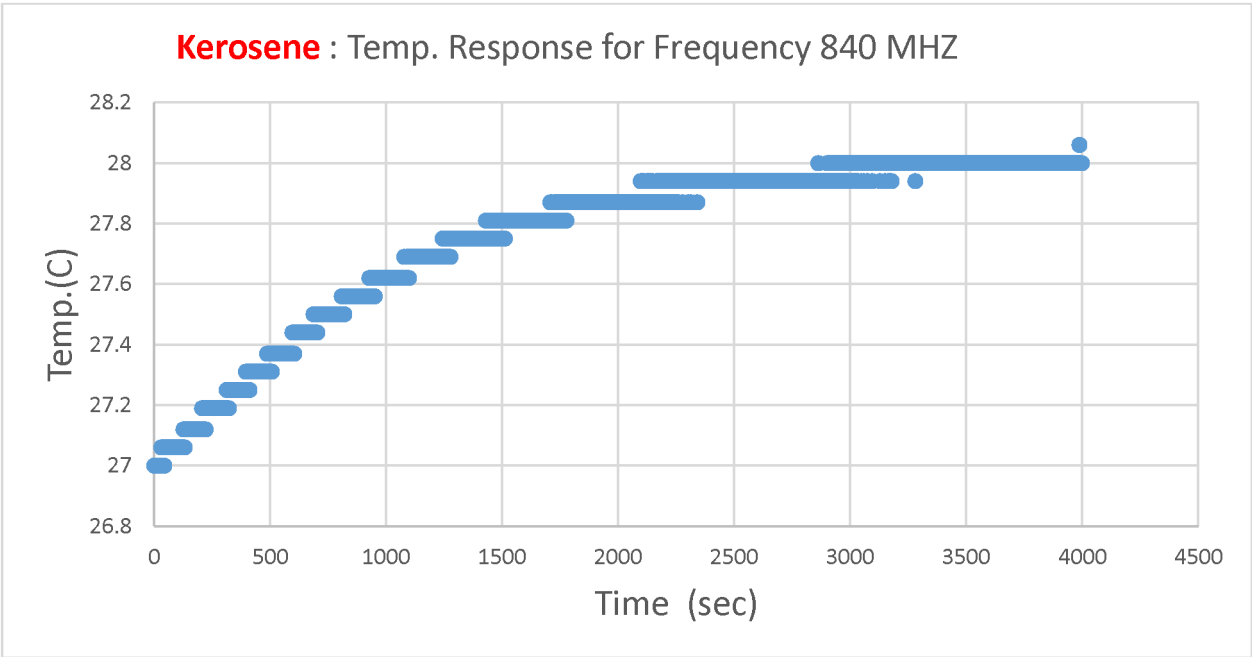
Graph 1.21—Relationship Between Temperature Measurements and Radiation Frequency at 122 Mhz for Kerosene



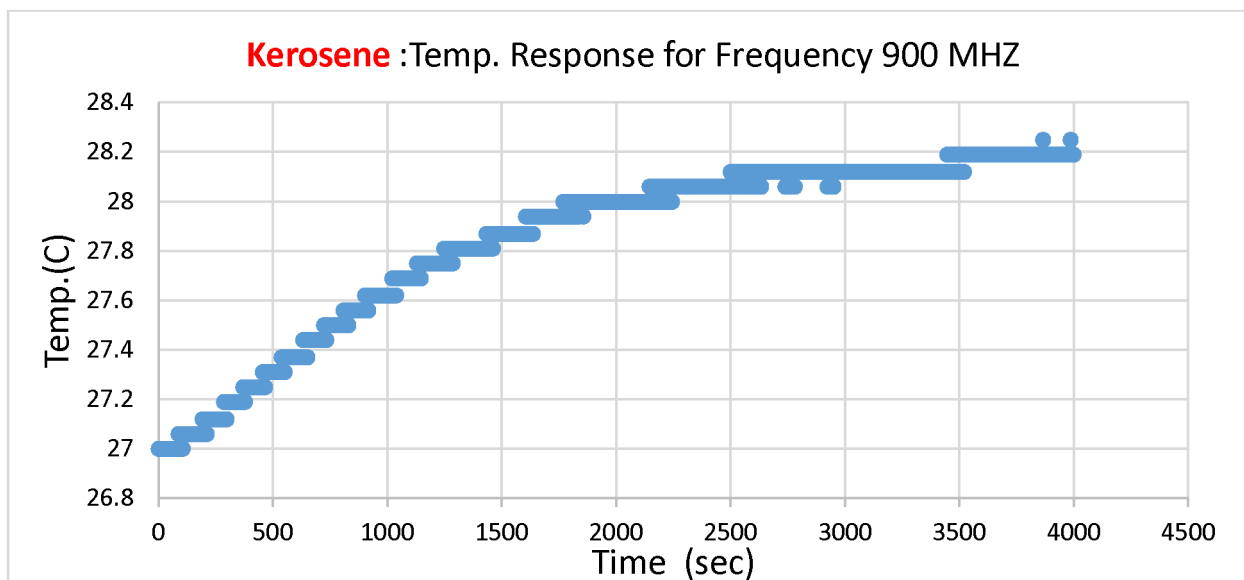
Graph 1.:22—Relationship Between Temperature Measurements and Radiation Frequency at 150 Mhz for Kerosene



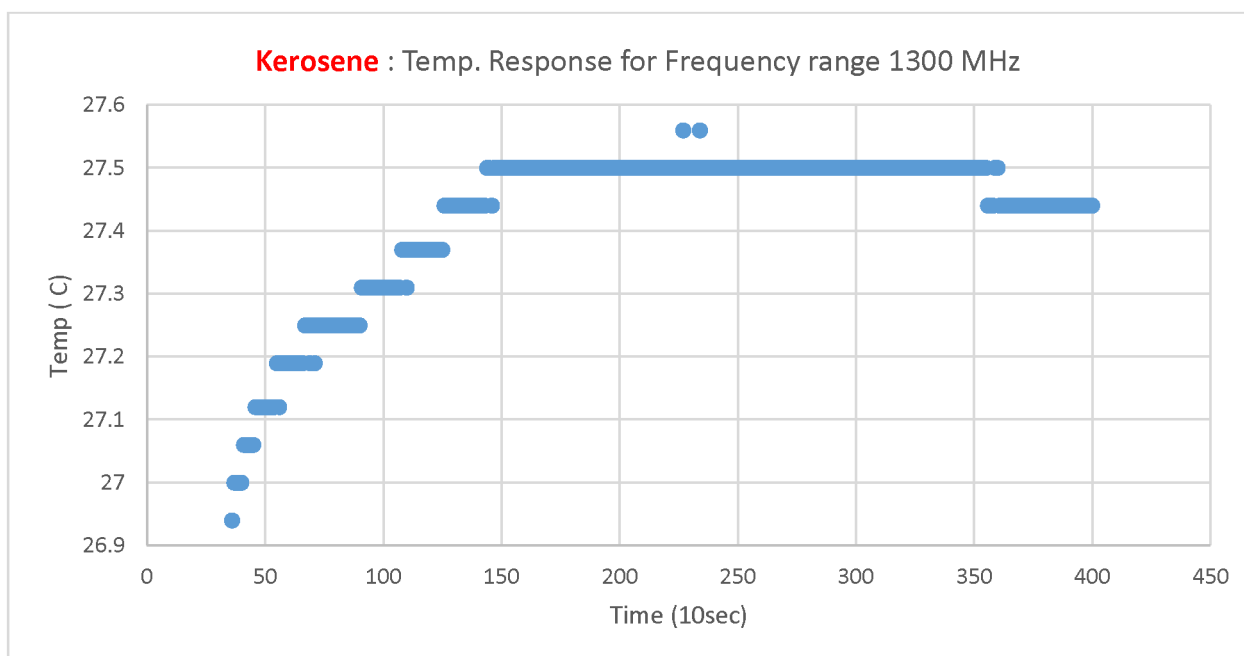
Graph 1.23—Relationship Between Temperature Measurements and Radiation Frequency at 630 Mhz for Kerosene



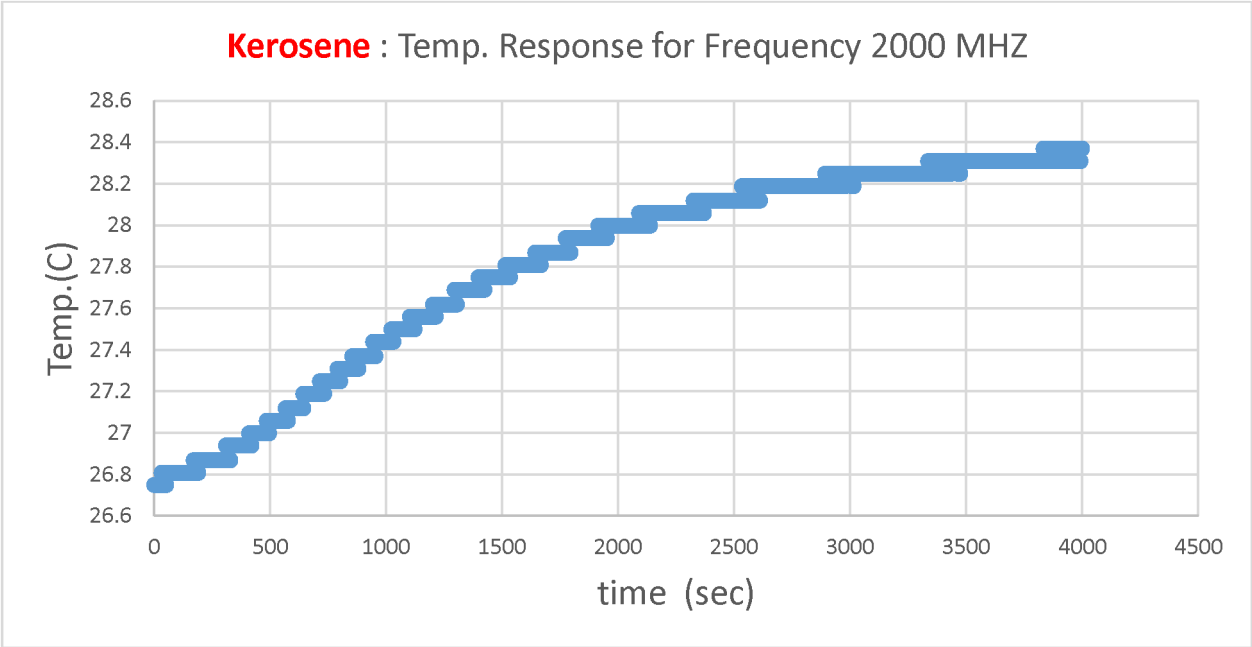
Graph 1.24—Relationship Between Temperature Measurements and Radiation Frequency at 840 Mhz for Kerosene



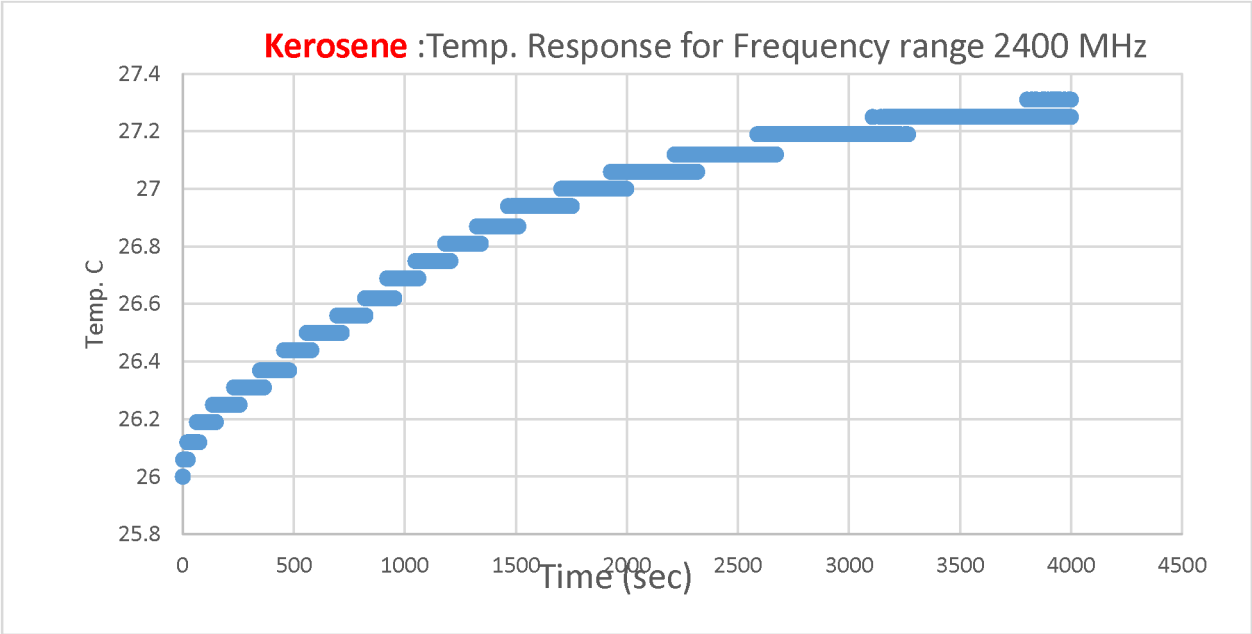
Graph 1.25—Relationship Between Temperature Measurements and Radiation Frequency at 900 Mhz for Kerosene



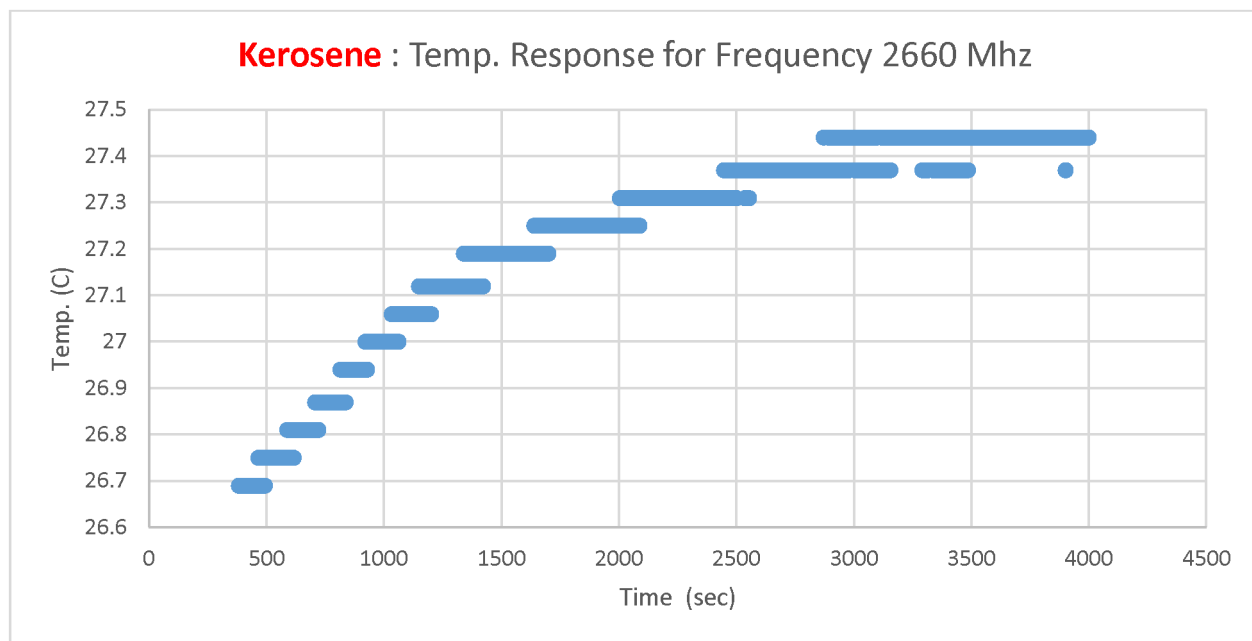
Graph 1.26—Relationship Between Temperature Measurements and Radiation Frequency at 1300 Mhz for Kerosene



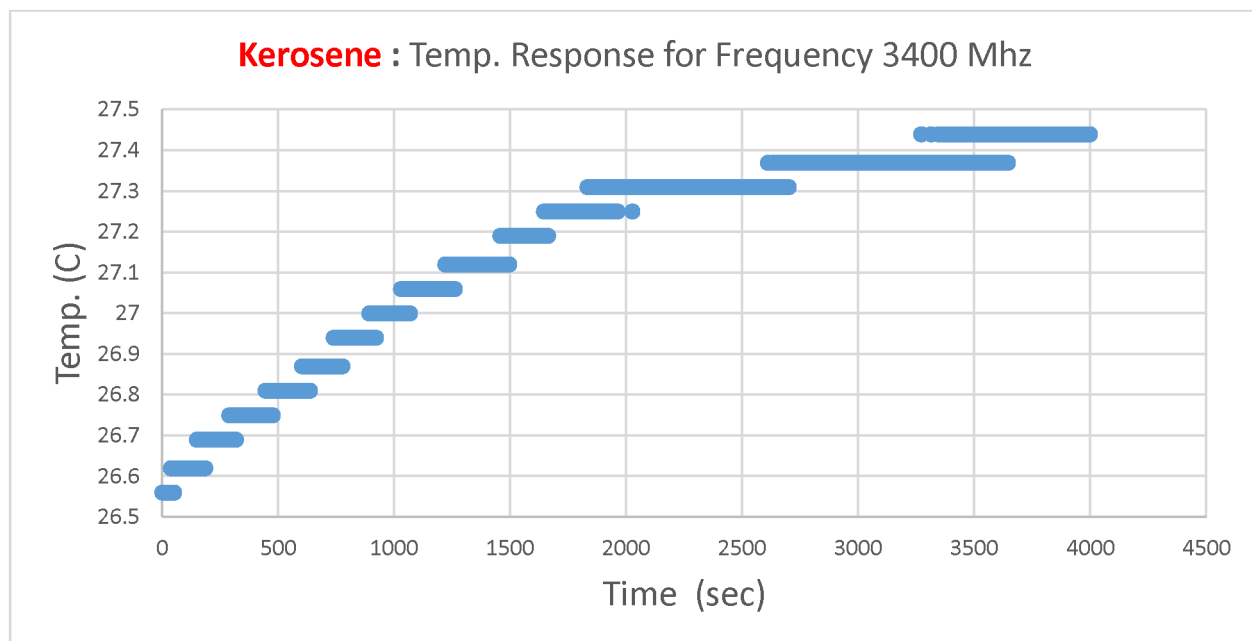
Graph 1.27—Relationship Between Temperature Measurements and Radiation Frequency at 2000 Mhz for Kerosene



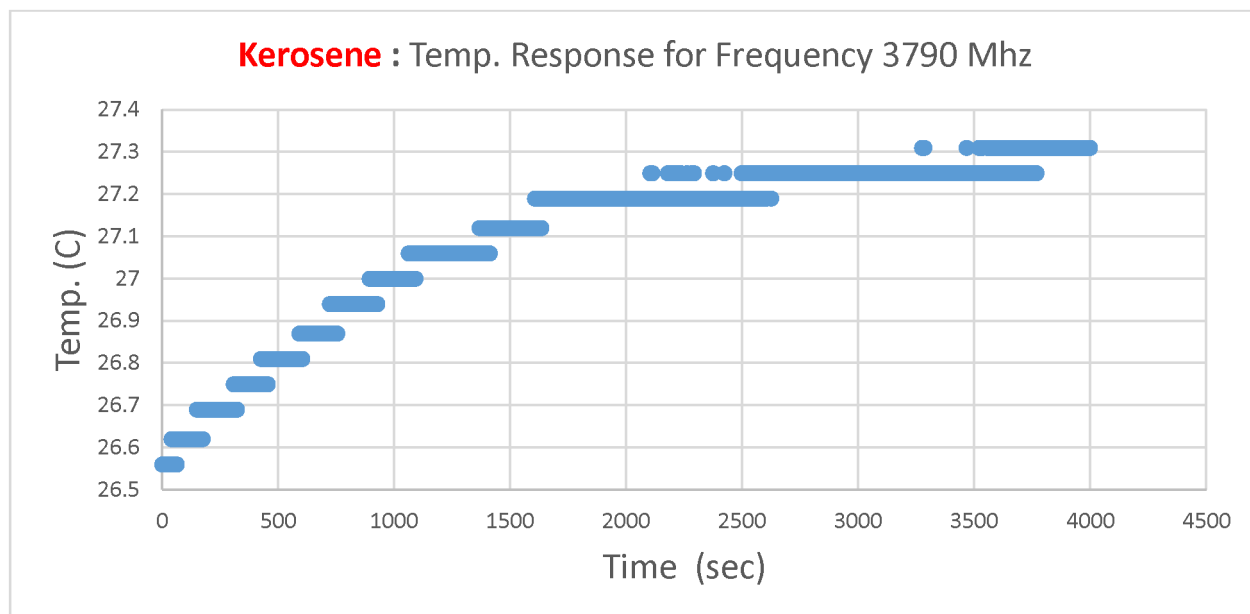
Graph 1.28—Relationship Between Temperature Measurements and Radiation Frequency at 2400 Mhz for Kerosene



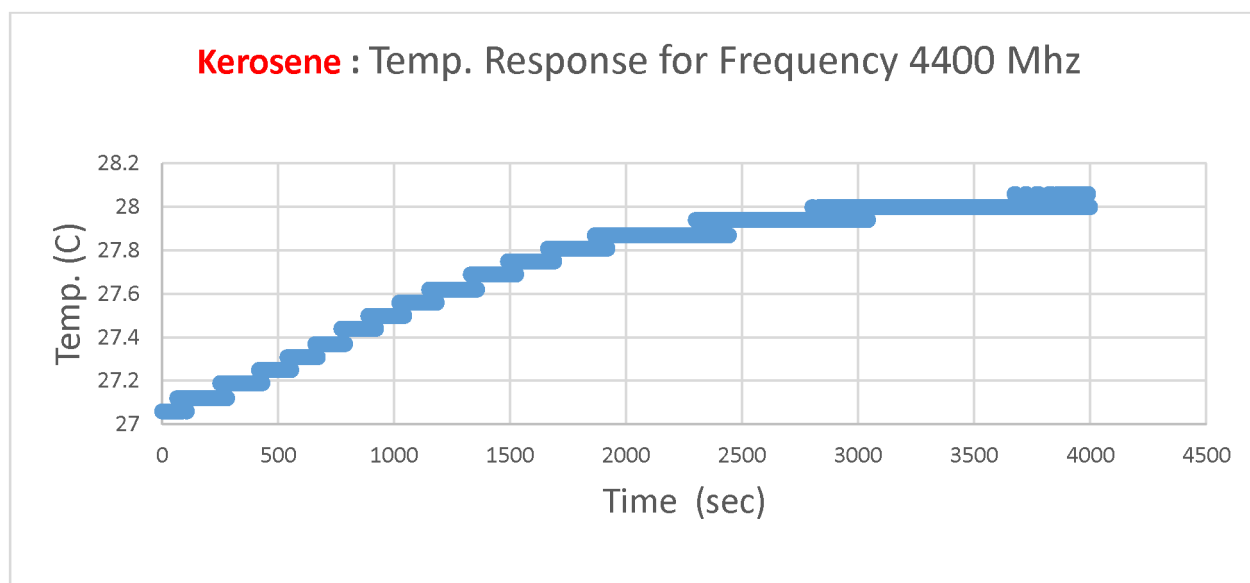
Graph 1.29—Relationship Between Temperature Measurements and Radiation Frequency at 2660 Mhz for Kerosene



Graph 1.30—Relationship Between Temperature Measurements and Radiation Frequency at 3400 Mhz for Kerosene

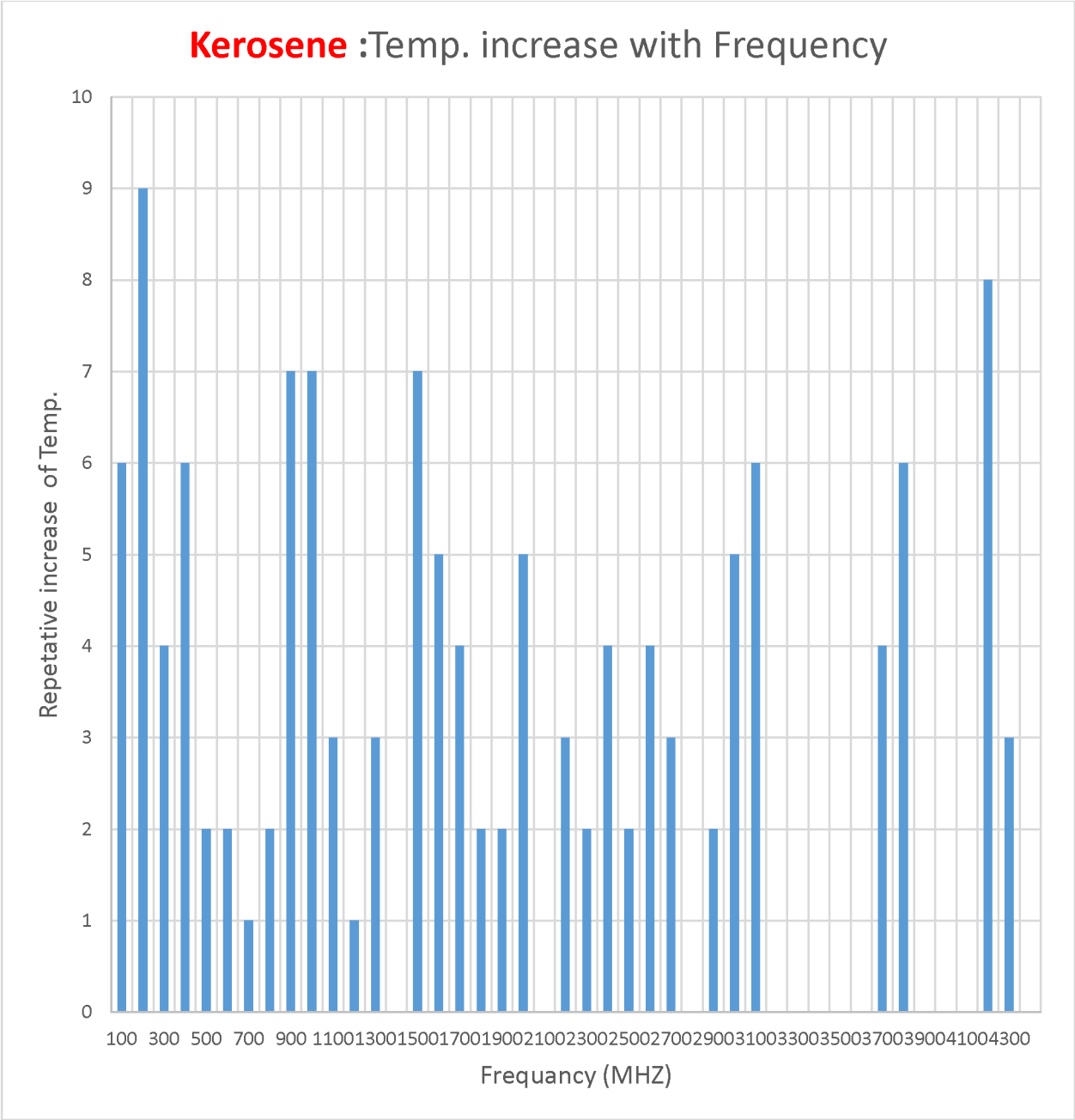


Graph 1.31—Relationship Between Temperature Measurements and Radiation Frequency at 3790 Mhz for Kerosene



Graph 1.32—Relationship Between Temperature Measurements and Radiation Frequency at 4400 Mhz for Kerosene

Repeated temperature increases and their associated frequencies Chart for Kerosene.



Graph 1.33—Relationship Between Frequency and Temperature Rise Repetition for Kerosene

Table 1.4—Measurements of Temperature Differences Over Time to Determine Kerosene Resonance Frequency

**Kerosene Comparison of energy consumption Table
for heating 1 cm³ of Kerosene by 1C° at different frequencies**

volume :14 cm³

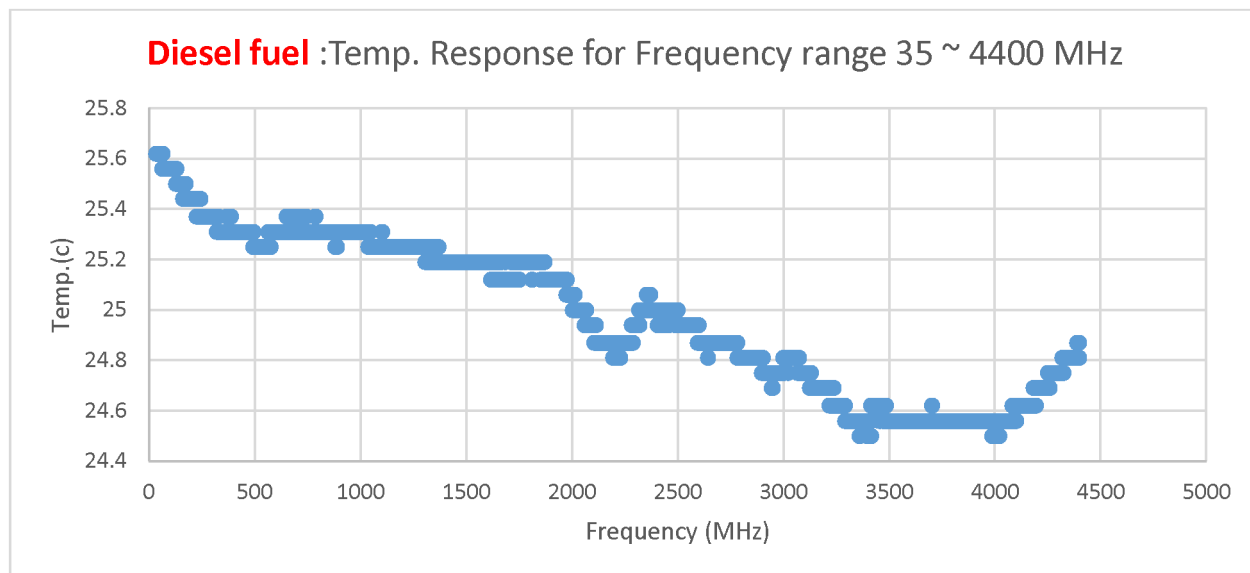
O/P power of pulse generator Board: 3.2 mWatt /hr.

No.	Acting frequency (MHZ)	MAX. heat Temp (C°)	MIN. heat Temp (C°)	Δ Temp. (C°)	Δ Time (sec)	Consumption power (mWatt)	Consumption power (mWatt) to heat 21 cm ³ by 1C°	Consumption power (mWatt) to heat 1 cm ³ by 1C°
1	65	29.75	28.75	1	4000	3.56	3.56	0.17
2	122	27.37	26.87	0.5	4000	3.56	7.11	0.34
3	150	29.94	28.69	1.25	4000	3.56	2.84	0.14
4	630	28.06	27.37	0.69	4000	3.56	5.15	0.25
5	840	28.06	27	1.06	4000	3.56	3.35	0.16
6	900	28.25	27	1.25	4000	3.56	2.84	0.14
7	1300	27.56	26.94	0.62	4000	3.56	5.73	0.27
8	2000	28.37	26.75	1.62	4000	3.56	2.19	0.10
9	2400	27.31	26	1.31	4000	3.56	2.71	0.13
10	2660	27.44	26.69	0.75	4000	3.56	4.74	0.23
11	3400	27.5	26.56	0.94	4000	3.56	3.78	0.18
12	3790	27.31	26.69	0.62	4000	3.56	5.73	0.27
13	4400	28.06	27.06	1	4000	3.56	3.56	0.17

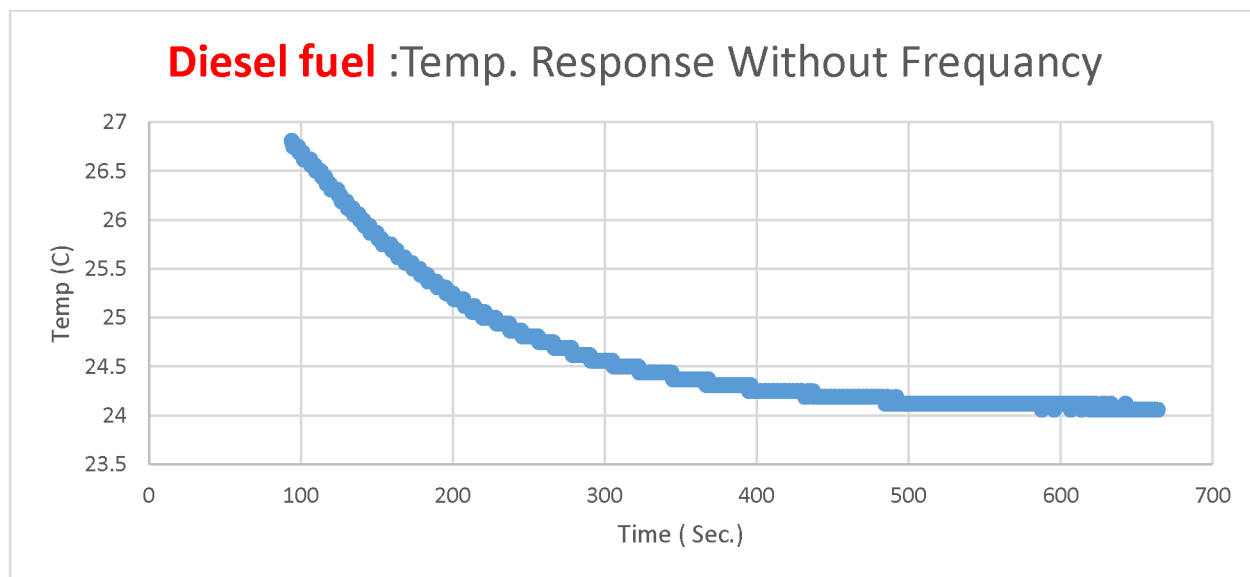
As result of above: Kerosene resonance frequency is **2000 MHz**

As result of above: Kerosene resonance frequency is 2000 MHz
Experimental Study for Identify Diesel Fuel Resonance Frequency.

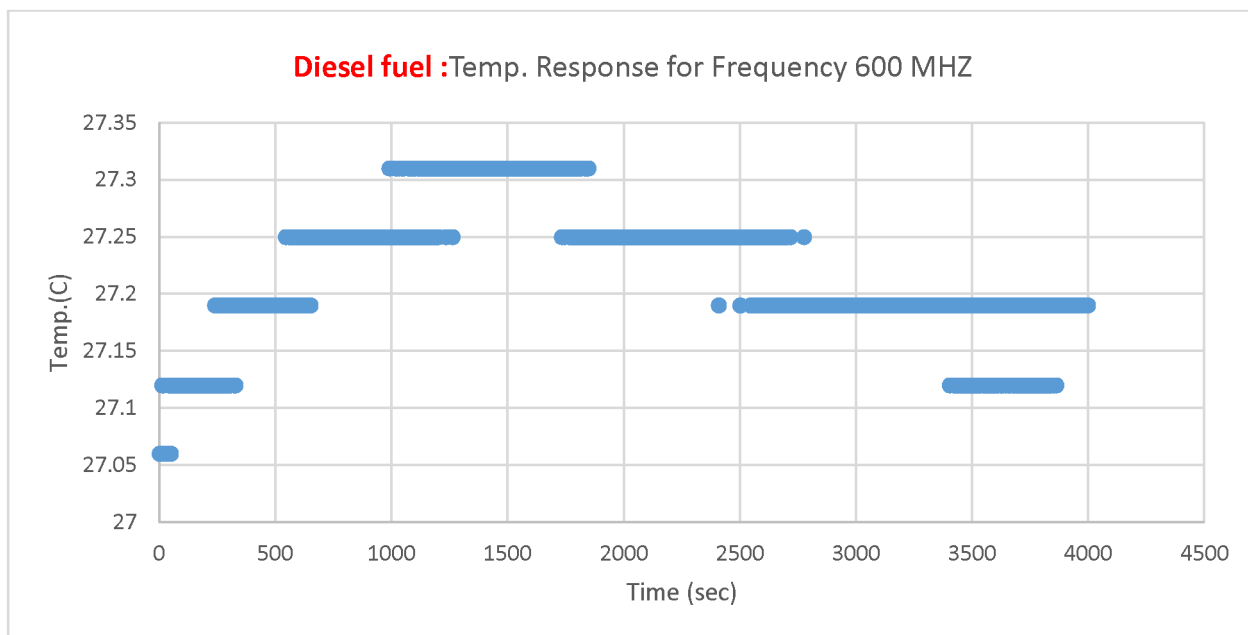
Charts Indicate the Relationship Between Radiated Frequency and The Recorded Temperature for Diesel Fuel.



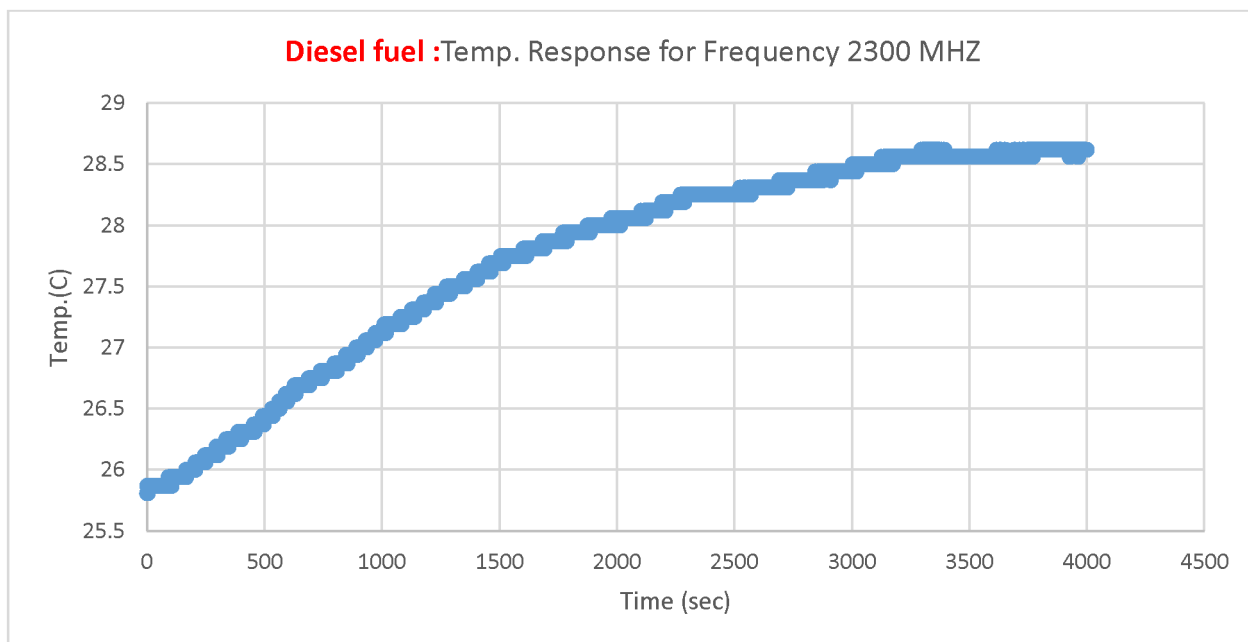
Graph 1.34—Relationship Between Temperature Measurements and Radiation Frequencies from 35 Mhz to 4400 Mhz for Diesel Fuel.



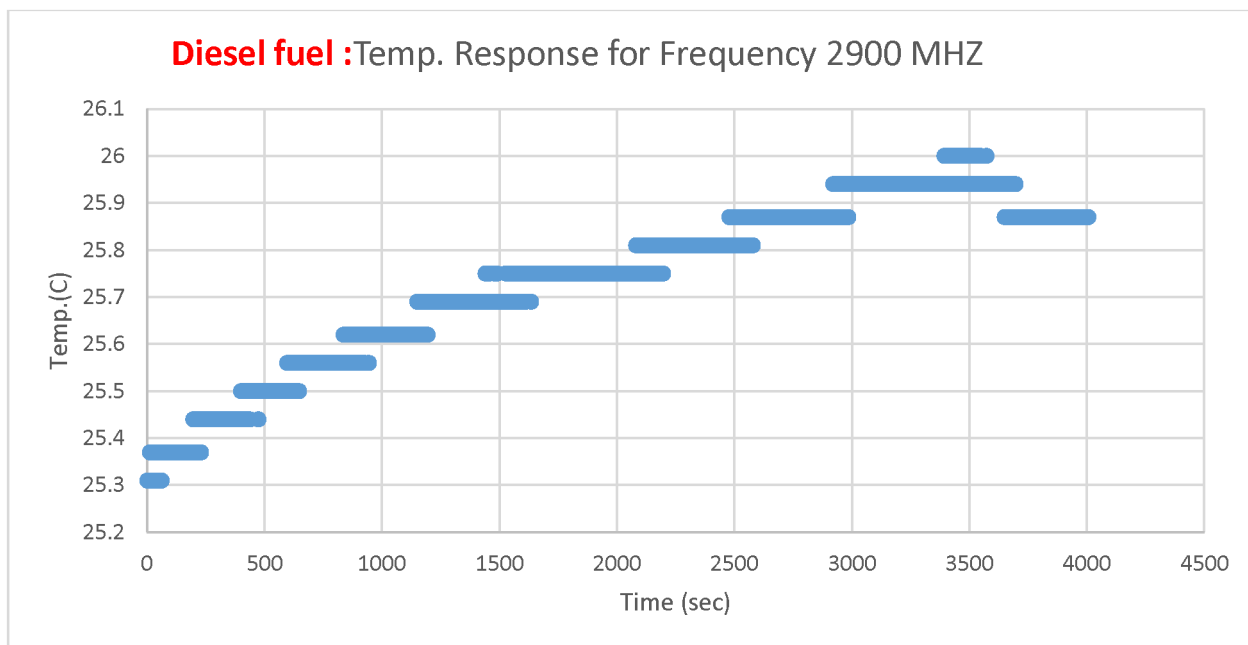
Graph 1.35—Relationship Between Temperature Measurements and Time without radiation for Diesel Fuel.



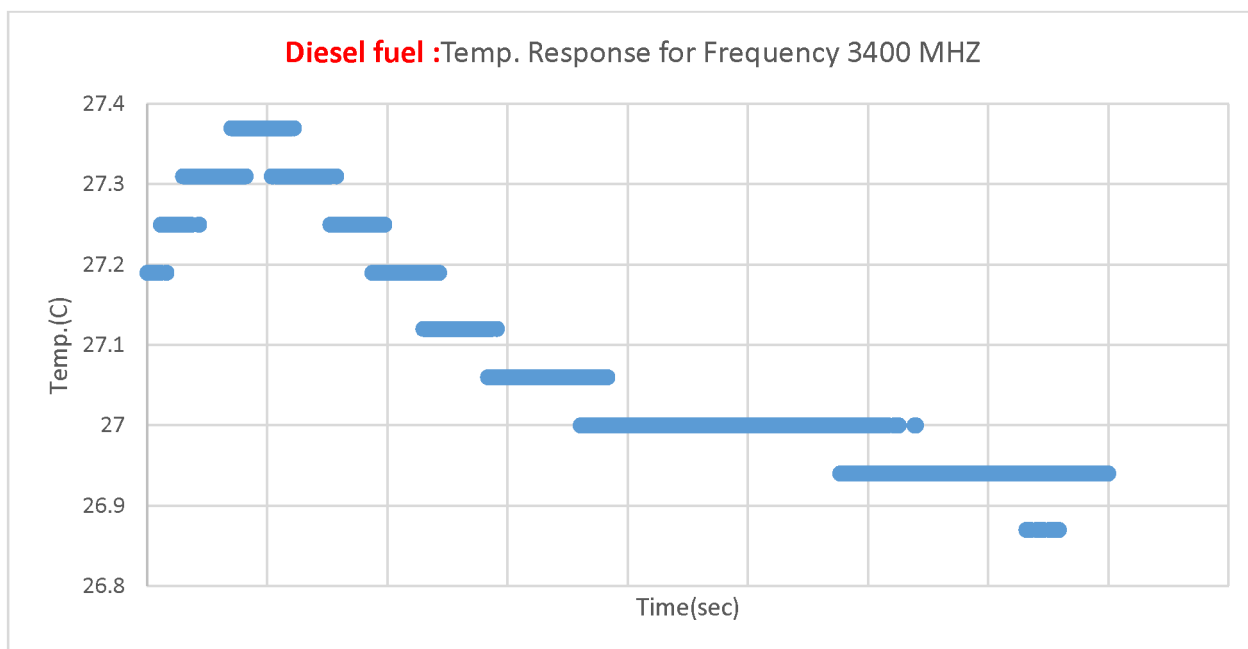
Graph 1.36—Relationship Between Temperature Measurements and Radiation Frequency at 600 Mhz for Diesel Fuel.



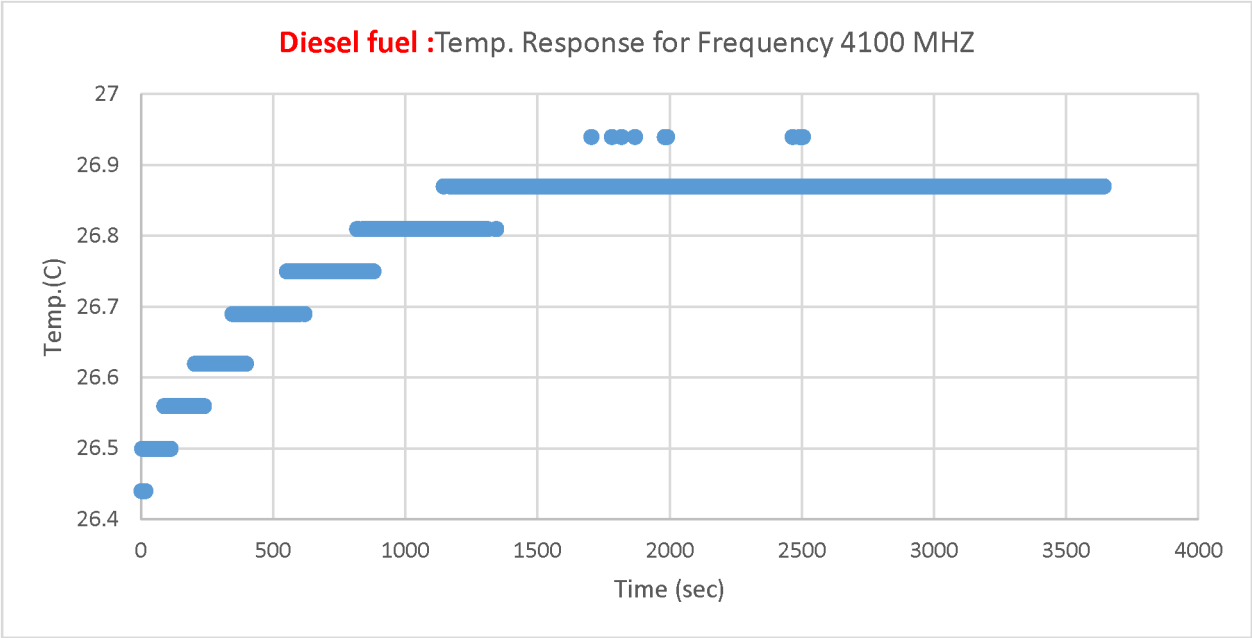
Graph 1.37—Relationship Between Temperature Measurements and Radiation Frequency at 2300 Mhz for Diesel Fuel.



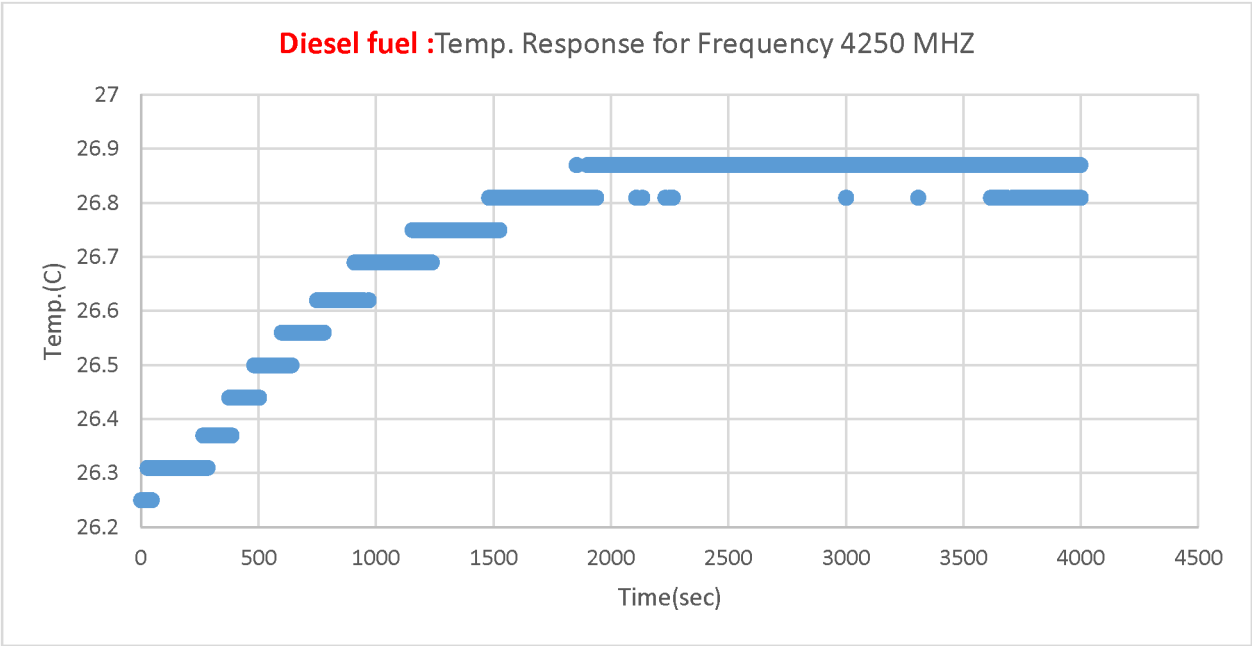
Graph 1.38—Relationship Between Temperature Measurements and Radiation Frequency at 2900 Mhz for Diesel Fuel.



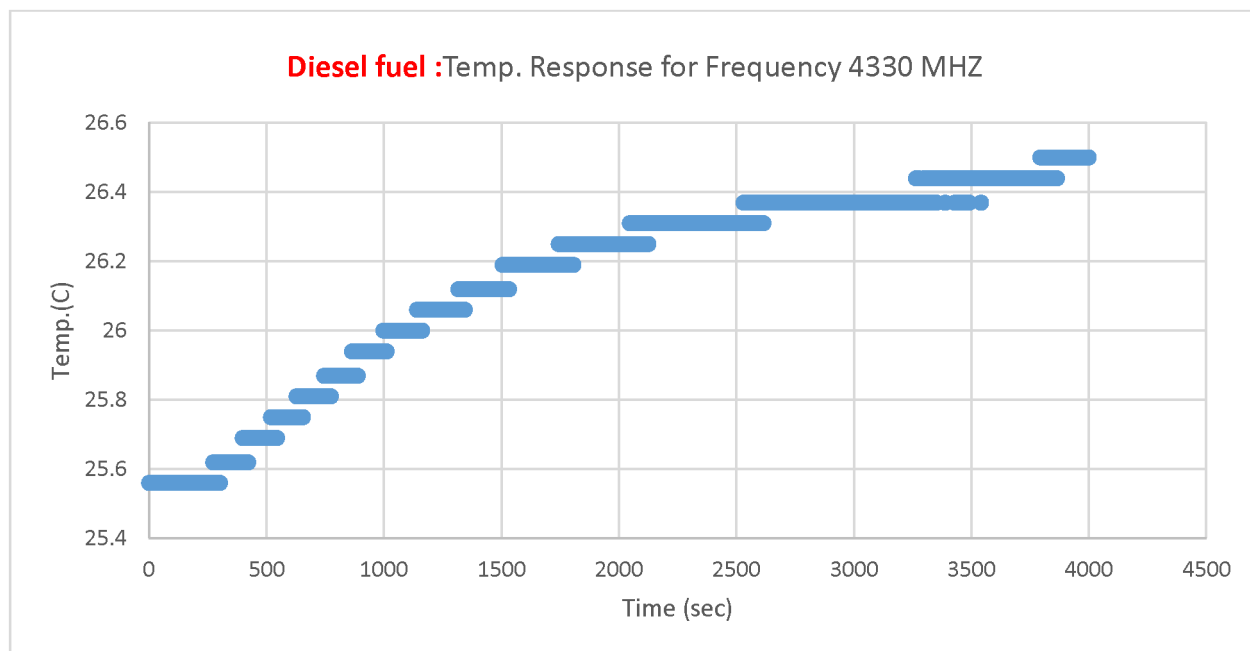
Graph 1.39—Relationship Between Temperature Measurements and Radiation Frequency at 3400 Mhz for Diesel Fuel.



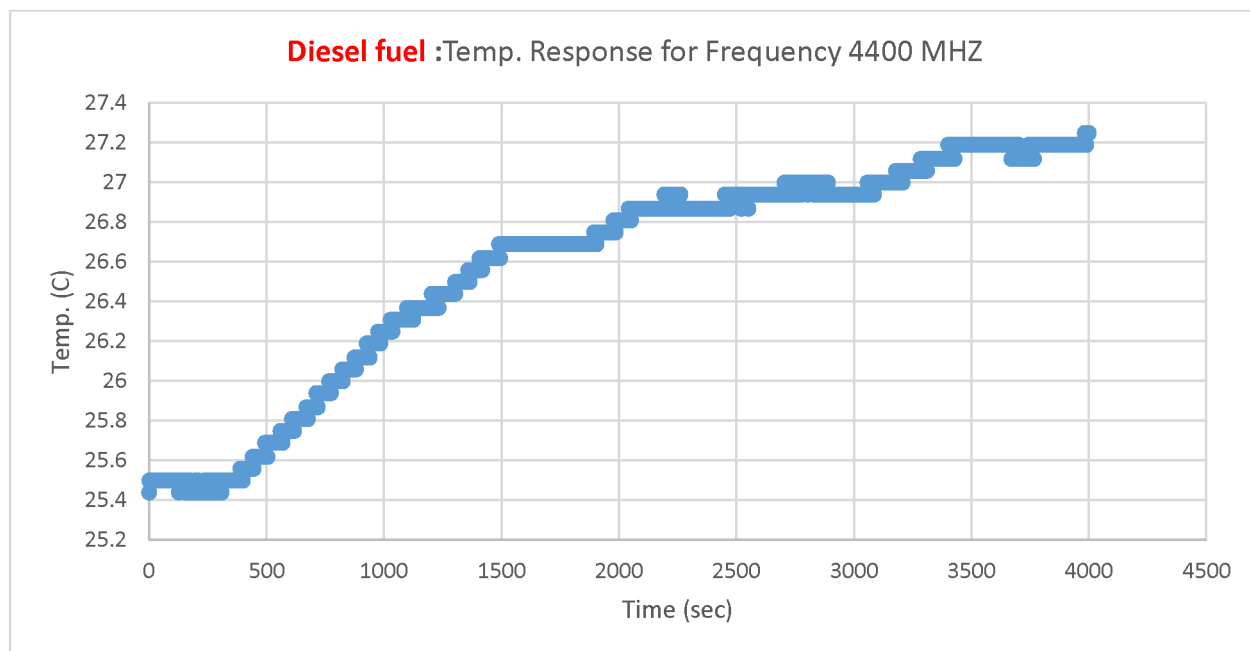
Graph 1.40—Relationship Between Temperature Measurements and Radiation Frequency at 4100 Mhz for Diesel Fuel.



Graph 1.41—Relationship Between Temperature Measurements and Radiation Frequency at 4250 Mhz for Diesel Fuel.

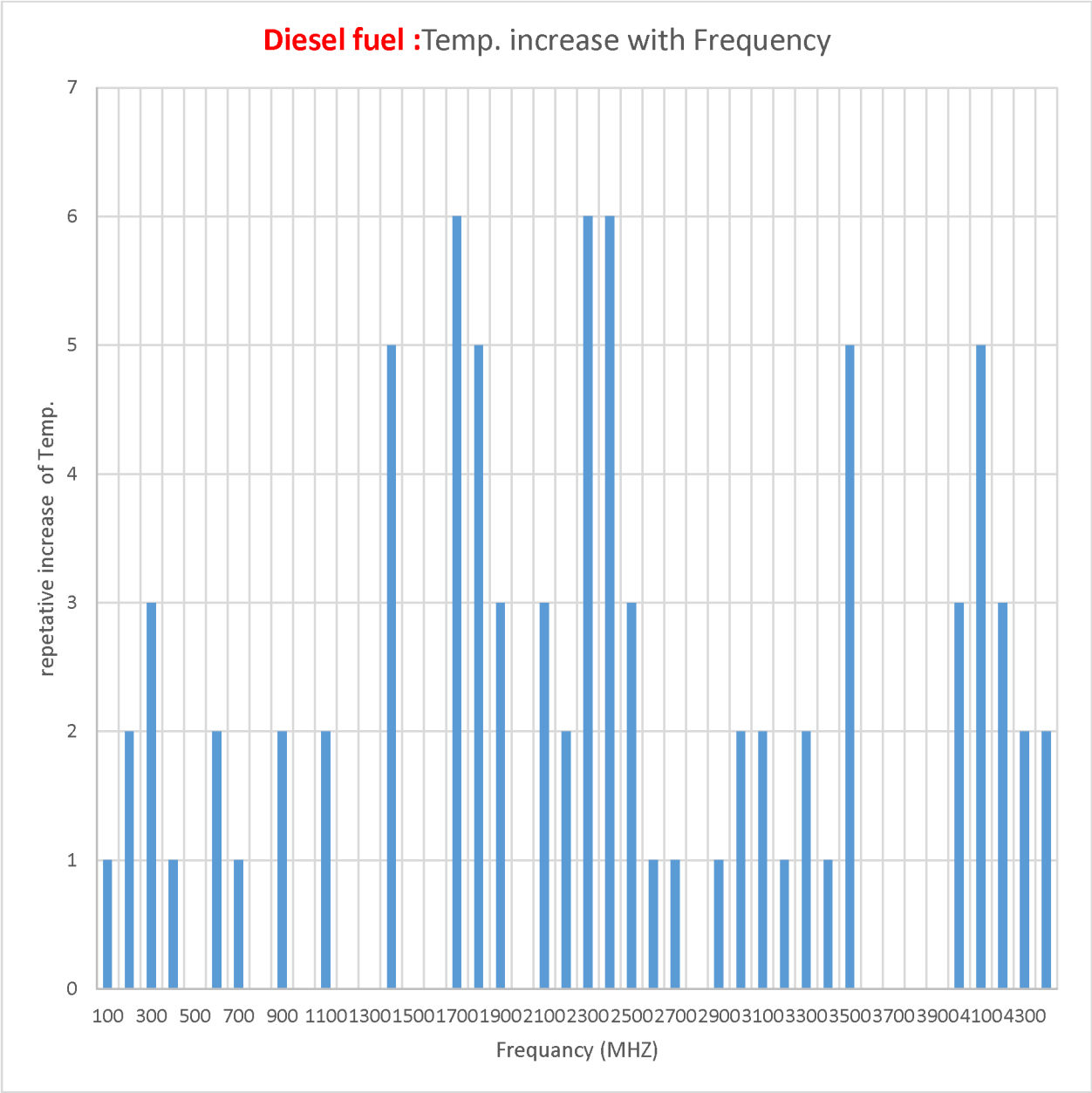


Graph 1.42—Relationship Between Temperature Measurements and Radiation Frequency at 4330 Mhz for Diesel Fuel.



Graph 1.43—Relationship Between Temperature Measurements and Radiation Frequency at 4400 Mhz for Diesel Fuel.

Repeated temperature increases and their associated frequencies Chart for Diesel Fuel.



Graph 1.44—Relationship Between Frequency and Temperature Rise Repetition for Diesel Fuel.

Table 1.5—Measurements of Temperature Differences Over Time to Determine Diesel Fuel Resonance Frequency

**Diesel Fuel Comparison of energy consumption Table
for heating 1 cm³ of Diesel Fuel by 1C° at different frequencies**

volume :14 cm³

O/P power of pulse generator Board: 3.2 mWatt /hr.

No.	Acting frequency (MHZ)	MAX. heat Temp (C°)	MIN. heat Temp (C°)	Δ Temp. (C°)	Δ Time (sec)	Consumption power (mWatt)	Consumption power (mWatt) to heat 21 cm ³ by 1C°	Consumption power (mWatt) to heat 1 cm ³ by 1C°
1	600	27.31	27.06	0.25	4000	3.56	14.22	1.02
2	2300	28.62	25.81	2.81	4000	3.56	1.27	0.09
3	2900	25.87	25.31	0.56	4000	3.56	6.35	0.45
4	3400	27.37	27.2	0.17	4000	3.56	20.92	1.49
5	4100	26.94	26.44	0.5	3643	3.24	6.48	0.46
6	4250	26.87	26.25	0.62	4000	3.56	5.73	0.41
7	4330	26.56	25.56	1	4000	3.56	3.56	0.25
8	4400	27.25	25.44	1.81	4000	3.56	1.96	0.14

As result of above: Diesel Fuel resonance frequency is **2300 MHz**

As result of above: Diesel Fuel resonance frequency is 2300 MHz